Geometric coherence of single-cell CRISPR perturbations reveals regulatory architecture and predicts cellular stress

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Abstract

Genome engineering has achieved sequence-level precision, yet predicting the transcriptomic state a cell will occupy after perturbation remains open. Single-cell CRISPR screens measure how far cells move, but effect magnitude ignores whether the cells move together. We introduce Shesha perturbation stability (S_p), which quantifies directional coherence as the mean cosine similarity between individual cell shift vectors and the mean perturbation direction. Across five CRISPR datasets (2,200+ perturbations), stability correlates with magnitude (Spearman $\rho = 0.75$–0.97), but discordant cases expose regulatory architecture: pleiotropic regulators such as CEBPA pay a "geometric tax," producing large but incoherent shifts, while lineage-specific factors such as KLF1 produce coordinated responses. S_p and Song et al.'s perturbation-response score (PS) share partial overlap ($\rho_{\text{partial}} = +0.51$ after controlling for magnitude), but S_p provides significant incremental prediction of UPR pathway activation beyond both PS and magnitude ($p < 10^{-18}$). In a split-half reproducibility assay, S_p predicts directional reproducibility beyond magnitude ($\rho_{\text{partial}} = +0.384$) while PS does not ($\rho_{\text{partial}} = -0.193$), with the advantage consistent across all magnitude strata and both datasets. Geometric instability is independently associated with UPR activation across four datasets. S_p is implemented in the open-source `shesha-geometry` Python package.

Introduction

The capacity to precisely edit genomes has outpaced our ability to predict the cellular consequences. CRISPR-Cas9 and its derivatives enable targeted modifications with unprecedented sequence-level accuracy (Doudna and Charpentier 2014; Jiang and Doudna 2017; Jinek et al. 2012), yet a cell can be edited exactly as intended and still drift toward an unintended fate. This gap between genetic precision and phenotypic predictability reflects three classes of failure that share a common feature. Off-target effects introduce unintended edits at genomically similar sites. On-target edits can trigger large deletions or chromothripsis invisible to standard sequencing (Kosicki, Tomberg, and Bradley 2018; Leibowitz et al. 2021). Most fundamentally, phenotypic heterogeneity is increasingly recognized as a defining challenge: two cells carrying the exact same edit often behave differently, one differentiating, one remaining stem-like (Replogle et al. 2022; Weinreb et al. 2020). The variation reflects the initial position of each cell on the state manifold and the local geometry of that landscape.

These failures occur in cell state space, not sequence space. Current evaluation frameworks measure the syntax of the edit: indel rates, off-target cleavage, and sequence fidelity (Brinkman et al. 2014; Tsai et al. 2014). They answer the engineer's question: was the code changed correctly? They do not answer the biologist's question: is the resulting state stable? We have mastered the syntax of the genome (Doudna and Charpentier 2014; Jiang and Doudna 2017; Jinek et al. 2012). We remain largely blind to its semantics.

To resolve this gap, we must pivot from a sequence-centric view of perturbation biology to a geometric one. The conceptual foundation already exists in developmental mechanics: the epigenetic landscape. When Conrad Waddington depicted cell development as a ball rolling down an undulating surface (Slack 2002; Waddington 1957), he was not merely offering an illustration. He was describing the topology of a dynamical system (Ferrell 2012). In this view, valleys are not metaphors; they are attractor basins, stable regions of state space where regulatory networks minimize the system's quasi-potential energy (Enver et al. 2009; Fard et al. 2016; S. Huang 2009; Wang et al. 2011). Ridges separating valleys represent unstable intermediates where small perturbations can redirect trajectories. Gene regulatory networks are optimized not merely for specific expression patterns but for the stability of those patterns under perturbation (Kitano 2004; Siegal and Bergman 2002).

Modern single-cell genomics has made this topology directly measurable (Rand et al. 2021). We can now observe thousands of cells responding to the same genetic perturbation and ask not only how far they moved from their unperturbed state but how they moved relative to one another (Nadig et al. 2025; Norman et al. 2019; Replogle et al. 2022). Yet the standard analytical framework for single-cell CRISPR screens reduces this rich geometric information to a single summary: effect magnitude, the distance between the mean perturbed and control expression profiles. Recent work has begun to characterize within-perturbation heterogeneity from complementary angles. Nadig et al. (Nadig et al. 2025) decomposed perturbation effects into shared and gene-specific components through transcriptome-wide differential expression analysis, revealing structure in how perturbations redistribute gene expression across the transcriptome. Song et al. (Song et al. 2025) introduced a perturbation-response score (PS) that estimates the strength of the perturbation effect for each individual cell, identifying which cells within a population responded strongly. Harmonized benchmarking resources (Peidli et al. 2024) have further enabled systematic comparison across screens. What remains unaddressed is the directional coherence of the population-level response: among the cells that did respond, did they move in the same direction? Two perturbations with identical magnitude can produce qualitatively different outcomes, one driving cells coherently along a shared transcriptomic trajectory, the other scattering them across expression space. Standard dimensionality reduction techniques (PCA, UMAP) compound this problem by projecting the high-dimensional state manifold onto flat coordinates, erasing the curvature that distinguishes deep attractors from shallow ridges (McInnes et al. 2018; Moon et al. 2019; Tsuyuzaki et al. 2020; Zhou and Sharpee 2021). Two cell populations that appear phenotypically similar in a reduced projection may occupy positions on the manifold separated by high energetic barriers.

Here we introduce a geometric stability metric, *Shesha*, that quantifies the directional coherence of single-cell perturbation responses. For each perturbation, we compute the shift vector from the control centroid for every perturbed cell, then measure the mean cosine similarity between these individual shift vectors and the mean perturbation direction. This score, which we term Shesha perturbation stability (S_p), captures whether cells respond to a genetic intervention by moving together (high S_p, coherent) or scattering (low S_p, incoherent). The metric adapts the principle of geometric self-consistency from a general representational stability framework (Raju 2026a,b) to the specific context of perturbation biology.

We validate perturbation stability across five single-cell CRISPR datasets (Adamson et al. 2016; Dixit et al. 2016; Norman et al. 2019; Papalexi et al. 2021; Replogle et al. 2022) spanning activation (CRISPRa), interference (CRISPRi), and pooled screens, comprising over 2,200 perturbations. Stability correlates strongly with effect magnitude across all datasets (Spearman $\rho = 0.75$–0.97), but the cases where the two metrics decouple are the most informative: pleiotropic master regulators such as CEBPA produce large but geometrically incoherent shifts, while lineage-specific factors such as KLF1 produce tightly coordinated responses. This "geometric tax" on pleiotropy emerges without supervision and distinguishes regulatory architecture from the data alone. Under nonlinear (LOESS) residual correction (Cleveland 1979; Cleveland and Devlin 1988), the most discordant perturbations in the genome-scale Replogle screen are spliceosome components and ESCRT membrane-remodeling factors, multi-subunit complexes whose disruption produces heterogeneous cellular responses. S_p captures information complementary to Song et al.'s perturbation-response score: after controlling for magnitude, the two metrics share partial overlap ($\rho = +0.51$), yet S_p provides significant incremental prediction of UPR pathway activation beyond both magnitude and PS. In a split-half reproducibility assay, S_p predicts which perturbation effects are directionally reproducible, with the advantage over magnitude consistent across all effect-size strata. After controlling for effect size, geometric instability is independently associated with activation of the unfolded protein response across all four testable datasets. The magnitude-stability relationship persists in nonlinear foundation model embeddings (scGPT; Cui et al. 2024), confirming it is a property of biological state space rather than an artifact of linear projection.

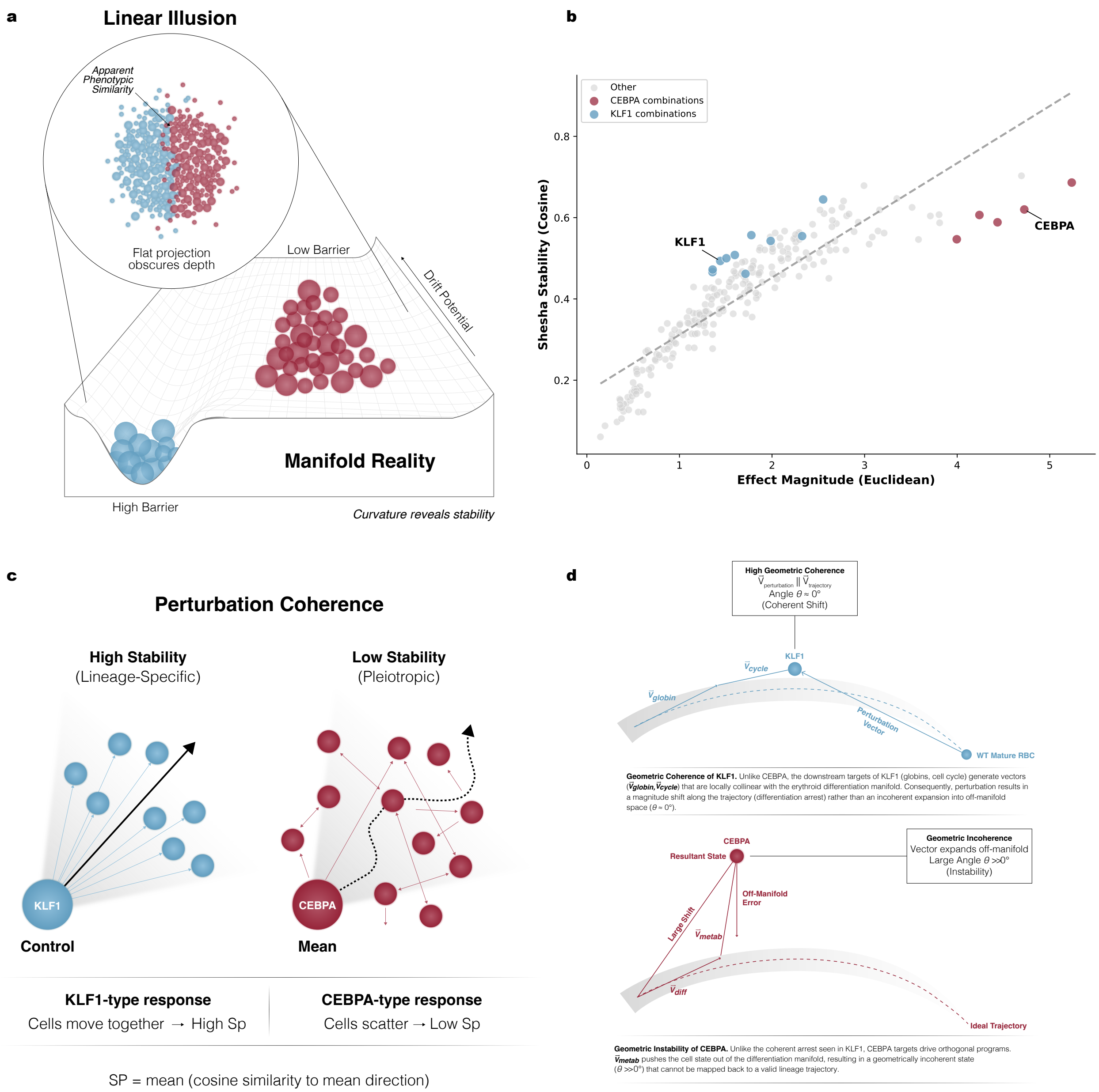

Figure 1: **The Geometric Tax: linear metrics obscure biological stability.** (**a**) Standard dimensionality reduction projects high-dimensional cell states onto a flat plane (Linear Illusion, inset), where two populations (blue, red) appear to overlap, suggesting similar phenotypes. Mapping these populations onto the underlying biological manifold (Manifold Reality) reveals distinct stability properties invisible to linear projections. The blue population occupies a deep valley (high barrier), representing a robust cell state resistant to perturbation. The red population sits on a shallow ridge (low barrier), representing an unstable state prone to drift. This stability difference constitutes the Geometric Tax of engineering cells into non-native configurations. (**b**) Stability versus magnitude for all 236 perturbations in Norman et al. (2019). CEBPA and its combinations (red) cluster below the regression line, indicating lower stability relative to their effect magnitude. KLF1 and its combinations (blue) cluster above. Dashed line: linear fit ($\rho = 0.953$). Note: the dashed line shows a linear fit; LOESS-corrected discordance rankings, which account for the nonlinearity of the magnitude-stability relationship at low magnitudes, differ for some high-magnitude perturbations. Under LOESS correction, CEBPA+JUN drops from linear rank 1 to LOESS rank 172, while KLF1 remains the most concordant single-gene perturbation under both methods (linear rank 236, LOESS rank 235; SI Appendix, Table S7). (**c**) Geometric stability quantified through perturbation coherence. High-stability perturbations (left, e.g., KLF1) produce shift vectors that align coherently, indicating cells move together along a shared trajectory toward the mean direction (solid arrow). Low-stability perturbations (right, e.g., CEBPA) scatter cells in divergent directions despite similar magnitude shifts, with the mean direction (dashed arc) representing dispersed cellular responses. The Shesha stability score (Sp) captures this distinction as the mean cosine similarity between individual shift vectors and the population mean. Together, panels a and b demonstrate how manifold curvature, invisible to linear projections, determines whether perturbations produce stable or fragile cellular states. (**d**) Geometric mechanism of perturbation coherence, illustrated schematically. Top: a lineage-specific perturbation (KLF1-type) produces shift vectors collinear with the differentiation manifold, resulting in coherent movement along an established trajectory (high S_p). Bottom: a pleiotropic perturbation (CEBPA-type) activates competing downstream programs whose shift vectors point in divergent directions, scattering cells off-manifold (low S_p).

Materials and Methods

Datasets

Five single-cell CRISPR perturbation datasets were analyzed (Table S1): Norman et al. 2019 (CRISPRa, K562, $n = 236$ perturbations including 105 single-gene and 131 combinatorial) (Norman et al. 2019); Adamson et al. 2016 (CRISPRi, K562, $n = 8$) (Adamson et al. 2016); Dixit et al. 2016 (CRISPRi, BMDCs, $n = 153$) (Dixit et al. 2016); Papalexi et al. 2021 (pooled CRISPR screen, THP-1, $n = 25$) (Papalexi et al. 2021); and Replogle et al. 2022 (genome-scale CRISPRi, K562, $n = 1{,}832$ perturbations after filtering for ≥ 50 cells per perturbation) (Replogle et al. 2022). All datasets were accessed via pertpy (v1.0.4) (Heumos et al. 2025).

Preprocessing

Each dataset was preprocessed independently to prevent cross-dataset batch effects. The pipeline consisted of: quality filtering (cells with fewer than 100 detected genes removed), library-size normalization (`scanpy.pp.normalize_total`; default parameters for Norman, Dixit, and Papalexi; `target_sum=1e4` for Replogle and Adamson), log transformation (`scanpy.pp.log1p`), selection of the top 2,000 highly variable genes (`scanpy.pp.highly_variable_genes`), and PCA with 50 components (`scanpy.tl.pca`). All downstream stability and magnitude computations were performed on the 50-dimensional PCA embedding. Robustness to PCA dimensionality (10, 20, 30, 50, 100 components) is confirmed in **SI Appendix, PCA Dimensionality Ablation**.

Control group identification

Control group assignment used a multi-stage matching protocol to accommodate heterogeneous labeling conventions across datasets. Labels were first matched case-insensitively against known control terms ("control", "ctrl", "non-targeting", "NT", "unperturbed"), then by delimiter-aware regex for short tokens, and finally by substring matching for embedded keywords. Dataset-specific handling: Replogle labels containing "non-targeting" or beginning with "chr" were assigned to control, and "pos_control" labels were removed; Papalexi non-targeting guides (NTg1–NTg7) were pooled into a single control group (2,386 cells); Norman cells labeled "control" served as the control group. Full details are provided in **SI Appendix, Control Group Identification**.

Quantifying geometric stability of perturbations

For each perturbation p applied to n_p cells, we computed the shift vector for each perturbed cell as the difference between its PCA coordinates and the control centroid. The mean perturbation direction is the average of these shift vectors. Shesha perturbation stability (S_p) is defined as the mean cosine similarity between individual cell shift vectors and this mean direction:

$$S_p = \frac{1}{n_p} \sum_{i=1}^{n_p} \frac{\mathbf{d}_i \cdot \bar{\mathbf{d}}}{\|\mathbf{d}_i\| \, \|\bar{\mathbf{d}}\|} \tag{1}$$

where $\mathbf{d}_i = \mathbf{x}_i - \boldsymbol{\mu}_{\text{ctrl}}$ is the shift vector for cell i and $\bar{\mathbf{d}} = \frac{1}{n_p} \sum_i \mathbf{d}_i$ is the mean perturbation direction. A perturbation with S_p near 1 drives all cells in the same direction (high coherence); S_p near 0 indicates scattering across expression space. Effect magnitude (M_p) is the Euclidean norm of the mean shift vector, $M_p = \|\bar{\mathbf{d}}\|$.

The metric adapts the principle of geometric self-consistency from the Shesha representational stability framework (Raju 2026a,b) to perturbation biology. Robustness to distance metric choice (Euclidean, Mahalanobis, k-NN) is confirmed in **SI Appendix, Distance Metric Robustness**. Perturbations with fewer than 50 cells (10 for Dixit) were excluded; the Adamson dataset retains all 8 perturbations despite wide bootstrap confidence intervals.

Discordance

To identify perturbations where magnitude and stability diverge, we computed discordance as the residual from the magnitude-stability relationship. Because this relationship is nonlinear at low magnitudes (where signal-to-noise constrains coherence), we used locally weighted scatterplot smoothing (LOESS, bandwidth fraction = 0.3) rather than ordinary least-squares regression. Positive discordance indicates lower stability than predicted for a given magnitude (below the curve); negative discordance indicates higher stability than predicted (above the curve). Linear and rank-based residual methods are compared in **SI Appendix, Nonlinear Discordance Comparison**; biological conclusions are robust to method choice, though specific gene rankings shift for high-magnitude perturbations.

Functional diversity of differentially expressed genes

To test whether geometric incoherence reflects the functional diversity of a perturbation's downstream effects, we computed the top-k differentially expressed genes for each perturbation in the Norman 2019 dataset, ranked by absolute log-fold change of mean expression versus control cells in log-normalized space. We tested $k \in \{25, 50, 100\}$ to assess sensitivity. Each gene list was submitted to g:Profiler (Kolberg et al. 2023) for GO Biological Process enrichment (organism: *H. sapiens*; significance threshold: FDR < 0.05; electronic annotations included). Functional diversity was defined as the number of distinct GO:BP terms with significant enrichment. Spearman correlations between discordance and functional diversity were computed for both linear and LOESS residual methods. Partial correlations controlling for effect magnitude were computed using Spearman partial correlation (Vallat 2018). The CEBP-family versus KLF1 comparison used a one-sided Mann-Whitney U test (alternative: CEBP > KLF1). All perturbations with zero enriched GO:BP terms were excluded from correlation analyses but retained for the group comparison.

Perturbation-response score comparison

To compare S_p against Song et al.'s perturbation-response score (Song et al. 2025), we computed PS using three approaches of increasing fidelity. Tier 1: mean per-cell Euclidean distance from the control centroid in PCA space. Tier 2: mean per-cell Mahalanobis distance, which re-weights PCA dimensions by the inverse control covariance matrix, accounting for axis-specific variance scales. Tier 3: a Python port of the scMAGeCK constrained-optimization algorithm (Yang et al. 2020), which estimates per-cell perturbation effects by regressing each cell's expression profile against differentially expressed marker genes identified for each perturbation. The centroid-based proxies (Tiers 1–2) correlate weakly with the scMAGeCK-derived PS ($\rho = 0.097$ and 0.149, respectively), indicating that per-cell distance from the control centroid does not approximate Song et al.'s score; all three tiers are reported to ensure transparency. Partial correlations between S_p and PS controlling for magnitude were computed via bootstrap resampling (10,000 iterations, seed 320). Incremental predictive power was assessed by computing the partial correlation of S_p with UPR pathway score after controlling for both magnitude and PS.

Comparison with pseudobulk heterogeneity metrics

To compare S_p-derived discordance with pseudobulk approaches, we computed η^2 (the proportion of PCA variance explained by perturbation identity) for each perturbation using one-way ANOVA across the first 50 principal components. Spearman correlations between η^2 and LOESS-residual discordance were computed for each dataset.

Split-half reproducibility assay

To test whether S_p predicts directional reproducibility of perturbation effects, we developed a split-half assay. For each perturbation in the Replogle ($n = 1{,}832$) and Norman ($n = 236$) datasets, cells were randomly partitioned, cells were randomly partitioned into two equal halves. The mean shift vector was computed for each half relative to the control centroid, and the cosine similarity between the two half shift vectors was recorded. This procedure was repeated for 50 independent random splits per perturbation, and the mean cosine similarity across splits served as the reproducibility estimate. Partial correlations between reproducibility and each predictor (S_p, PS, magnitude) were computed controlling for magnitude. To control for residual magnitude confounding, perturbations were stratified into magnitude-matched quartiles and the S_p advantage was assessed within each bin. The reproducibility advantage of S_p over PS was assessed across three PS implementations (Euclidean, Mahalanobis, and scMAGeCK) to ensure robustness to the PS estimator.

Pathway-level stress scoring

To assess the association between geometric stability and cellular stress, we computed composite pathway scores for four MSigDB Hallmark gene sets (Liberzon et al. 2015): Unfolded Protein Response (UPR), p53 Pathway, Apoptosis, and Reactive Oxygen Species (ROS). Pathway scores were computed using `scanpy.tl.score_genes` with curated gene lists

(72–78 overlapping genes for UPR, 32–51 for p53, 53–73 for Apoptosis, 45–55 for ROS per dataset) on the full normalized transcriptome before highly variable gene selection, then correlated with perturbation stability controlling for effect magnitude. Both raw Spearman correlations and partial correlations (controlling for magnitude) are reported, with Benjamini-Hochberg correction applied across the 16 pathway-dataset combinations.

At the individual marker level, four canonical stress response genes (DDIT3, ATF4, XBP1, HSPA5) were assessed using the same partial correlation framework. Quadrant depletion tests split perturbations at median stability and median stress score, testing whether the high-stability/high-stress quadrant was depleted relative to independence using a one-sided binomial test. Functional category stratification of the Replogle dataset grouped perturbations by GO molecular function annotations to assess whether the UPR-stability association was driven by any single gene class.

To test whether the CRISPRa null in Norman masked a within-stratum signal, perturbations were stratified into magnitude quartiles and HSPA5 expression was compared between high- and low-discordance groups within each bin using a one-sided Mann-Whitney U test.

scGPT validation

To test whether the magnitude-stability relationship is specific to linear PCA embeddings, we recomputed stability and magnitude in scGPT "Whole Human" pretrained embeddings (Cui et al. 2024) for three datasets (Norman, Dixit, Replogle). Raw counts (not log-normalized) were embedded using `embed_data()` with deterministic settings. Stability and magnitude were computed identically to the PCA pipeline. Full embedding protocol and reproducibility settings are provided in **SI Appendix, scGPT Validation Protocol**.

Statistical analysis

All confidence intervals were computed via bootstrap resampling (10,000 iterations, seed 320, percentile method). Cross-dataset generalization was assessed with a linear mixed-effects model (dataset as random effect; fixed effects: magnitude, spread, sample size; full specification in **SI Appendix, Mixed-Effects Model**). Partial correlations between stability and stress markers were computed controlling for effect magnitude. All p-values are two-sided unless otherwise noted.

Results

Stability tracks perturbation magnitude across modalities

Across all five datasets, perturbation stability correlates strongly and positively with effect magnitude (**Fig. 2**). The relationship is robust: Spearman correlations range from $\rho = 0.746$ in Dixit (95% CI $[0.641, 0.827]$) to $\rho = 0.985$ in Papalexi ($[0.939, 0.997]$), with the two largest datasets yielding $\rho = 0.953$ (Norman, $[0.934, 0.965]$) and $\rho = 0.970$ (Replogle, $[0.966, 0.972]$). Even Adamson, with only eight perturbations, shows $\rho = 0.929$ ($[0.407, 1.000]$). This consistency across CRISPRa and CRISPRi modalities, across cell types (K562, BMDCs, HeLa), and across screen scales (8 to 1,832 perturbations) indicates that the magnitude-stability relationship is a general property of perturbation geometry in single-cell expression space.

Table 1: Magnitude-stability correlation across five CRISPR datasets. Spearman ρ between effect magnitude (Euclidean distance) and perturbation stability (S_p, cosine coherence) in PCA space. Bootstrap 95% confidence intervals (10,000 resamples). Final row shows pooled (z-scored) results. These rank correlations are fit-independent; LOESS fits across all five datasets is provided in **SI Appendix, Table S5 and Fig. S1**.

Dataset	Modality	n	ρ	95% CI	p
Norman	CRISPRa	236	0.953	[0.934, 0.965]	$< 10^{-100}$
Adamson	CRISPRi	8	0.929	[0.407, 1.000]	$< 10^{-4}$
Dixit	CRISPRi	153	0.746	[0.641, 0.827]	$< 10^{-28}$
Papalexi	Pooled	25	0.985	[0.939, 0.997]	$< 10^{-100}$
Replogle	CRISPRi	1,832	0.970	[0.966, 0.972]	$< 10^{-19}$
Pooled	—	2,254	0.969	[0.965, 0.971]	$< 10^{-100}$

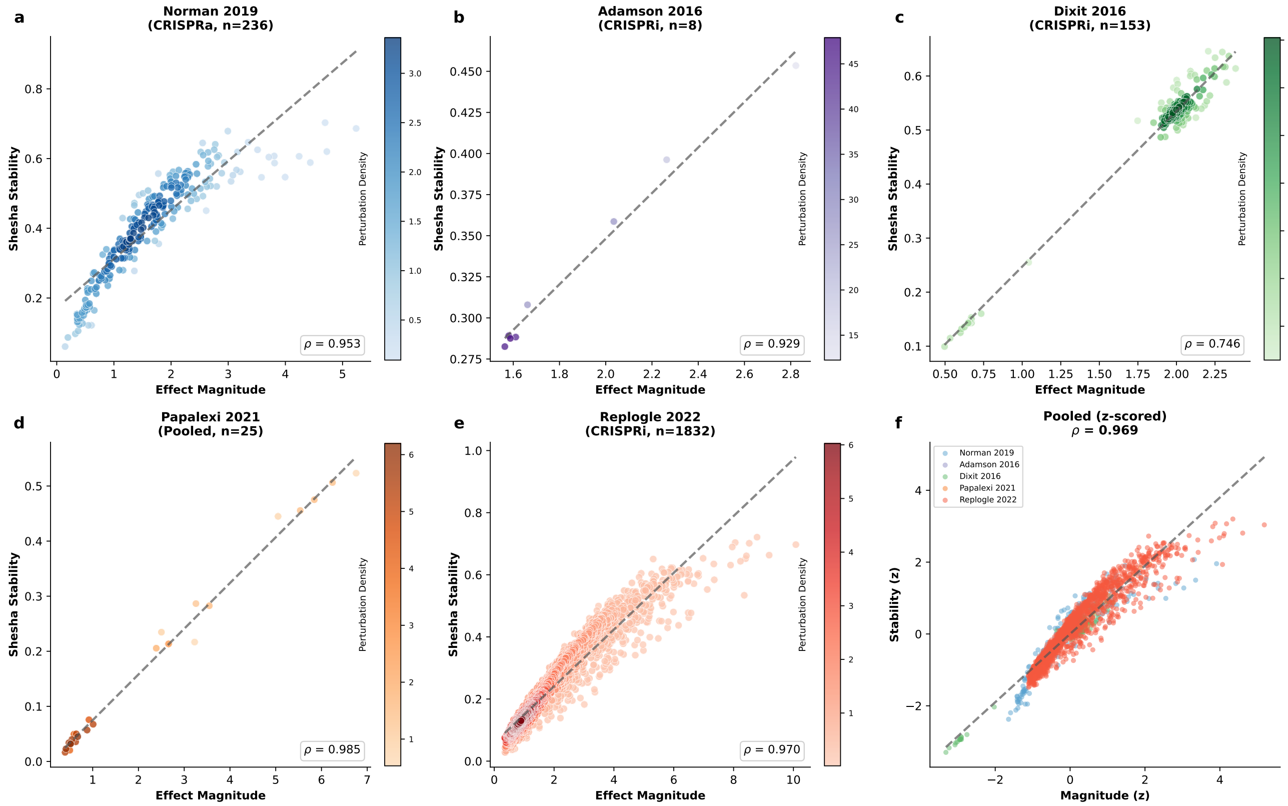

Figure 2: **Perturbation stability tracks effect magnitude across CRISPR modalities and cell types** (**a–e**) Effect magnitude (Euclidean distance, x-axis) vs perturbation stability (S_p, cosine coherence, y-axis) for each of five datasets: (**a**) Norman 2019 CRISPRa in K562 ($n = 236$, $\rho = 0.953$), (**b**) Adamson 2016 CRISPRi ($n = 8$, $\rho = 0.929$), (**c**) Dixit 2016 CRISPRi in BMDCs ($n = 153$, $\rho = 0.746$), (**d**) Papalexi 2021 pooled screen ($n = 25$, $\rho = 0.985$), (**e**) Replogle 2022 genome-scale CRISPRi in K562 ($n = 1{,}832$, $\rho = 0.970$). Dashed lines: linear regression. (**f**) Pooled cross-dataset scatter after z-score normalization within each dataset (calibrated $\rho = 0.968$, 95% CI $[0.965, 0.971]$). All Spearman correlations with bootstrap 95% CIs (10,000 resamples).

To confirm that this relationship generalizes beyond any single dataset, we z-scored magnitude and stability within each dataset and pooled all 2,254 perturbations. The calibrated cross-dataset correlation was $\rho = 0.968$ (95% CI $[0.965, 0.971]$; **Fig. 2f**). A linear mixed-effects model with dataset as a random effect confirmed that the dataset-level random-effect variance was near zero, and that magnitude was the dominant predictor of stability ($\beta = 0.168$, $[0.166, 0.170]$), accounting for approximately 11 times more variance than sample size ($\beta_{\text{n\_cells}} = -0.015$). The relationship was also robust to the choice of distance metric: Mahalanobis (whitened) and k-nearest neighbor matched controls produced consistent or stronger correlations (**SI Appendix, Robustness Analyses**).

The strength of this correlation has a straightforward geometric interpretation: perturbations that push cells further from the control state tend to do so more coherently, because a large mean shift requires that individual shift vectors share a common direction. The interesting biology lies not in this expected correlation but in the cases where it breaks down.

Magnitude and stability decouple for pleiotropic regulators

Although magnitude and stability are strongly correlated, they are not redundant. To identify perturbations where the two metrics diverge, we computed discordance as the residual from the magnitude-stability relationship. Because this relationship is nonlinear at low magnitudes (where signal-to-noise constrains coherence), we used locally weighted scatterplot smoothing (LOESS, bandwidth fraction = 0.3) rather than ordinary least-squares regression. Perturbations with large positive discordance produce large transcriptomic shifts but unexpectedly low geometric coherence; perturbations with large negative discordance are more coherent than their effect size would predict. Linear and rank-based residual methods are compared in **SI Appendix, Tables S7-S9**; the biological conclusions are robust to method choice, though specific gene rankings shift for high-magnitude perturbations where the nonlinearity is strongest.

In the Norman CRISPRa dataset, the discordance pattern depends on the residual method, illustrating why the nonlinear correction matters. Under linear residuals, the C/EBP transcription factor family dominates the discordant extreme: CEBPA+JUN (rank 1), CEBPA (rank 2), and CEBPA+CEBPB (rank 3) cluster below the linear regression line (**Fig. 1b**). These perturbations have the highest magnitudes in the dataset ($M_p = 8$–10), precisely the regime where the linear fit underpredicts stability. Under LOESS correction, the CEBPA discordance is substantially attenuated: CEBPA+JUN drops from rank 1 to rank 172 (with slightly negative discordance), and CEBPA solo drops from rank 2 to rank 64. Some CEBPA combinations remain moderately discordant (CEBPA+ZC3HAV1: LOESS rank 17; CEBPA+CEBPB: rank 36), indicating that the geometric tax on pleiotropy is real but smaller than the linear analysis suggested. The LOESS-corrected discordant extreme is instead occupied by moderate-magnitude perturbations: PLK4+STIL (mitotic kinase and centriole factor; rank 1), HES7 (Notch pathway oscillator; rank 2), and C3orf72+FOXL2 (rank 3) (**SI Appendix, Table S7**).

By contrast, KLF1 concordance is robust to the choice of method. KLF1 is the most concordant single-gene perturbation under both linear (rank 236) and LOESS (rank 235) residuals, and KLF1 combinations (KLF1+SET, DUSP9+KLF1, BAK1+KLF1) consistently occupy the concordant extreme. The erythroid-specific transcription factor whose targets are coordinated toward terminal red blood cell maturation (Miller and J J Bieker 1993; Siatecka and James J. Bieker 2011; Tallack and Perkins 2010) produces geometric coherence regardless of how discordance is computed, confirming that the lineage-specific coherence finding is not an artifact of the residual method.

The same pattern emerges independently in the Replogle genome-scale CRISPRi screen (n = 1,832 perturbations; **Fig. S2**) (Replogle et al. 2022), where the LOESS-corrected discordance analysis reveals a striking functional pattern among the most discordant genes. The top five are CHMP2A (ESCRT-III membrane remodeling (McCullough, Colf, and Sundquist 2013); discordance = 5.07), SF3B3 (U2 snRNP splicing factor; discordance = 4.83), SF3B2 (U2 snRNP splicing factor; discordance = 4.55), PSMD7 (26S proteasome regulatory subunit; discordance = 4.23), and CHMP3 (ESCRT-III membrane remodeling (McCullough, Colf, and Sundquist 2013); discordance = 4.20). The broader top ten includes additional spliceosome components (SMU1, CRNKL1, NSRP1), a COPI vesicle coat subunit (COPB2), and an ESCRT-I component (TSG101) (**Table S9**). These genes share a common structural feature: they encode subunits of large, essential molecular machines. When one subunit of a multi-protein complex is lost, the downstream consequences depend on the residual complex activity in each cell, the specific processing step that becomes rate-limiting, and the cell's current demand for that machinery. The result is heterogeneous cellular responses that register as geometric incoherence. At the concordant extreme, ribosome biogenesis factors (LSG1, ISG20L2, KRI1) produce tightly coherent shifts despite moderate effect sizes (Couté et al. 2008; Hedges, West, and Johnson 2005), consistent with the narrow functional scope of these genes.

Two observations clarify the interpretation. Low geometric stability is not a proxy for cell cycle arrest: BUB3 (spindle assembly checkpoint (Taylor, Ha, and McKeon 1998)) and CENPW (centromere protein (Hori et al. 2008)) both show low stability in Replogle, but BLVRB (biliverdin reductase (Wu et al. 2016)) shows high stability while cells continue cycling. Discordance quartile analysis reveals that the most discordant perturbations (Q4) have approximately three times the stability variance of the most concordant (Q1) in both Norman (SD = 0.202 vs 0.074) and Replogle (SD = 0.183 vs 0.097), while median cell counts are comparable across quartiles, ruling out a sample size confound. In every case, the shared feature of high-discordance perturbations is broad regulatory scope: multi-subunit complexes or transcription factors whose downstream targets span multiple functional programs.

To test whether geometric incoherence reflects functional diversity of downstream targets, we annotated the top 50 differentially expressed genes (ranked by absolute log-fold change versus control) for each Norman perturbation with GO Biological Process terms via g:Profiler (FDR < 0.05). CEBP-family perturbations showed significantly higher functional diversity than KLF1 combinations (mean 21.1 vs 9.6 distinct GO:BP terms; Mann-Whitney $U = 151$, $p = 0.013$), providing direct evidence that the geometric tax on CEBPA reflects its broad regulatory scope rather than a statistical artifact. However, LOESS-residual discordance did not correlate with functional diversity across all perturbations ($\rho = -0.119$, $p = 0.11$ at $k = 50$; sensitivity analysis at $k = 25$ and $k = 100$ in **SI Appendix, Table S16**), indicating that transcriptional target diversity is one source of geometric incoherence but not the only one. The Replogle analysis, where the most discordant perturbations are multi-subunit complex components rather than pleiotropic transcription factors, points to heterogeneous partial loss-of-function as a second, independent mechanism.

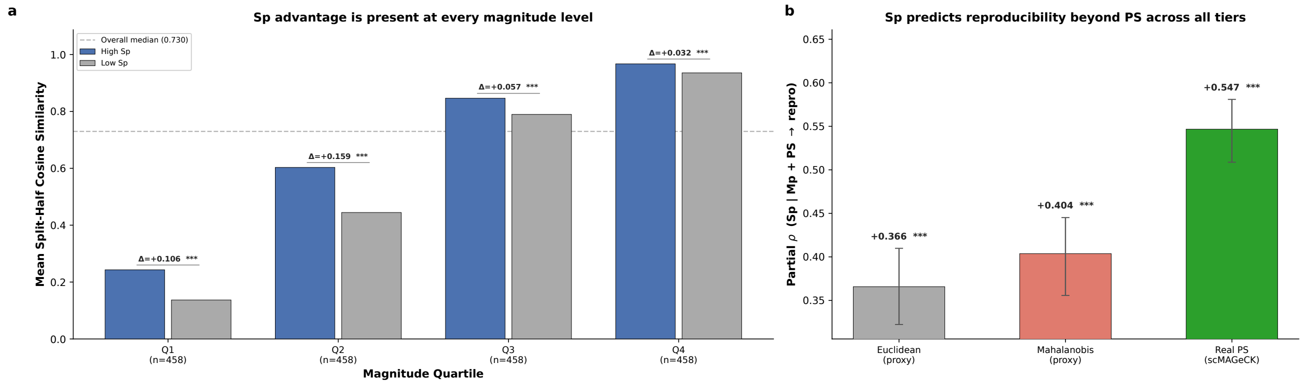

Figure 3: **Perturbation stability predicts directional reproducibility beyond both magnitude and perturbation-response scores.** (**a**) Magnitude-matched quartile analysis in Replogle 2022 ($n = 1{,}832$). Within each magnitude quartile ($n = 458$ per bin), perturbations are split at the median S_p value. High-S_p perturbations (blue) show greater split-half cosine similarity than low-S_p perturbations (gray) in all four bins (all $p < 10^{-9}$). The advantage is largest at low magnitudes (Q1–Q2), where hit prioritization is most challenging. Dashed line: overall median split-half cosine similarity. (**b**) Partial correlation of S_p with split-half reproducibility after controlling for both magnitude and PS, across three PS implementations. S_p retains significant incremental reproducibility prediction regardless of PS estimator (all $p < 0.001$), with the strongest effect when controlling for the real scMAGeCK PS ($\rho = +0.547$). Error bars: 95% bootstrap CIs.

Perturbation stability predicts screen hit reproducibility

A metric that captures geometric coherence should predict which perturbation effects are directionally reproducible: if perturbed cells move coherently, independent subsamples should recover the same shift direction. We tested this prediction using a split-half reproducibility assay. For each of the 1,832 perturbations in the Replogle dataset, we randomly partitioned cells into two equal halves, computed the mean shift vector for each half relative to the control centroid, and measured the cosine similarity between the two half-shift vectors. Averaging over 50 independent random splits per perturbation yielded a stable estimate of directional reproducibility for each gene knockout.

Perturbation stability and split-half reproducibility are expected to correlate, because both reflect the directional coherence of single-cell responses. The relevant question is not whether the correlation exists but whether S_p provides predictive information beyond what effect magnitude alone supplies. Magnitude dominates raw prediction of reproducibility (Spearman $\rho = 0.943$) because larger perturbation effects have higher signal-to-noise ratios regardless of their geometric structure. After controlling for magnitude, however, S_p retained a substantial positive partial correlation with split-half reproducibility (partial $\rho = +0.387$, 95% CI $[+0.345, +0.428]$, $p = 1.3 \times 10^{-66}$). All three implementations of Song et al.'s perturbation-response score showed the opposite pattern: negative partial correlations with reproducibility after magnitude control (Euclidean proxy: -0.249; Mahalanobis proxy: -0.184; scMAGeCK algorithm: -0.193, $p = 7.1 \times 10^{-17}$). At equivalent magnitude, perturbations with higher geometric coherence are more reproducible, while perturbations where individual cells respond more strongly are less so (**Fig. 3a**).

To verify that this relationship was not driven by residual magnitude confounding, we stratified perturbations into magnitude-matched quartiles and compared split-half cosine similarity between the upper and lower halves of the S_p distribution within each bin (**Fig. 3b**). High-S_p perturbations showed significantly greater directional reproducibility in every magnitude quartile: Q1 (lowest magnitude, $\Delta = +0.106$, $p = 1.3 \times 10^{-17}$), Q2 ($\Delta = +0.159$, $p = 4.1 \times 10^{-23}$), Q3 ($\Delta = +0.057$, $p = 3.3 \times 10^{-10}$), and Q4 (highest magnitude, $\Delta = +0.032$, $p = 3.2 \times 10^{-28}$). The effect was largest for low-magnitude perturbations (Q1 and Q2), precisely the regime where screen hit prioritization is most difficult and where geometric coherence provides the most information beyond effect size.

The same pattern replicated in the Norman CRISPRa dataset ($n = 236$). S_p showed an even stronger partial correlation with reproducibility (partial $\rho = +0.485$, $p = 2.6 \times 10^{-15}$), while the real PS again showed a negative partial correlation (-0.180, $p = 0.006$). High-S_p perturbations had greater split-half cosine similarity in all four magnitude quartiles, with the largest advantage at low magnitudes (Q1: $\Delta = +0.156$, $p = 5.8 \times 10^{-16}$).

These results establish a practical application for perturbation stability in CRISPR screen analysis. When two gene knockouts produce effects of similar magnitude, the one with higher S_p is more likely to yield a directionally reproducible phenotype. This advantage is robust across two datasets, two perturbation modalities, all magnitude strata, and all three

PS implementations tested (full benchmarking in **SI Appendix, Table S17**).

Perturbation stability captures information complementary to perturbation-response scores

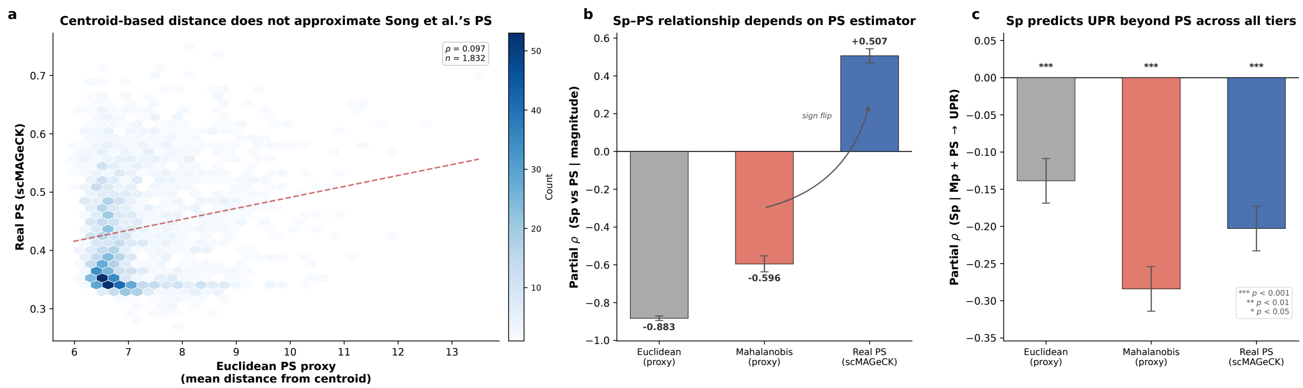

Figure 4: **Perturbation stability provides non-redundant stress prediction beyond perturbation-response scores.** (**a**) Centroid-based Euclidean distance proxy versus Song et al.'s algorithm-derived PS (scMAGeCK) for 1,832 Replogle perturbations. The two measures are essentially uncorrelated ($\rho = 0.097$), indicating that per-cell distance from the control centroid does not approximate the perturbation-response score. (**b**) Partial correlation between S_p and PS after controlling for effect magnitude, computed with three PS estimators. The relationship reverses from strongly negative (centroid-based proxies) to moderately positive (real PS), revealing that the proxy-based anticorrelation was a geometric artifact rather than a biological property. (**c**) Incremental prediction of UPR pathway activation: partial correlation of S_p with UPR score controlling for both magnitude and PS. S_p retains significant incremental prediction regardless of PS estimator (all $p < 10^{-9}$), confirming that directional coherence captures stress-relevant information beyond per-cell response strength.

The perturbation-response score (PS) introduced by Song et al. (Song et al. 2025) estimates the strength of each individual cell's response to a perturbation. We asked whether S_p and PS capture redundant or complementary information, comparing S_p against PS computed at three levels of fidelity: a centroid-based Euclidean distance proxy, a Mahalanobis distance proxy (re-weighted by control covariance), and a Python port of Song et al.'s scMAGeCK constrained-optimization algorithm (Yang et al. 2020) (**SI Appendix, Table S10**).

The centroid-based proxies correlate weakly with Song et al.'s algorithm-derived PS (Replogle: $\rho = 0.097$ for Euclidean, $\rho = 0.149$ for Mahalanobis), indicating that mean per-cell distance from the control centroid does not approximate the perturbation-response score. This distinction matters: the centroid-based proxies anticorrelate strongly with S_p after controlling for magnitude (Euclidean: partial $\rho = -0.883$; Mahalanobis: -0.596), but this anticorrelation reflects a geometric tautology rather than biological complementarity. Coherent movement concentrates cells near the mean shift vector, mechanically reducing per-cell distance from the centroid; the "complementarity" is between S_p and a noisy magnitude proxy, not between S_p and an independent measure of perturbation response.

With Song et al.'s algorithm-derived PS, the relationship reverses. After controlling for magnitude, S_p and the real PS are moderately positively correlated (partial $\rho = +0.507$, 95% CI $[+0.468, +0.544]$, $p < 10^{-120}$), sharing approximately 20% of their variance. Both metrics partially capture the same underlying signal (perturbation strength beyond simple magnitude), but each retains substantial unique information.

The critical test is whether S_p provides incremental prediction of cellular stress beyond what the real PS and magnitude together explain. It does: S_p retained a significant partial correlation with UPR pathway score after controlling for both magnitude and PS ($\rho = -0.203$, 95% CI $[-0.248, -0.156]$, $p = 1.9\times10^{-18}$; **Fig. 4b**). The real PS showed a weaker incremental contribution beyond S_p and magnitude ($\rho = -0.072$, $p = 2.1 \times 10^{-3}$). This asymmetry indicates that the directional coherence captured by S_p contains stress-relevant information that per-cell response strength does not fully account for. The result held across all three PS tiers: Euclidean ($\rho = -0.139$, $p < 10^{-9}$), Mahalanobis ($\rho = -0.284$, $p < 10^{-35}$), and real PS ($\rho = -0.203$, $p < 10^{-18}$), confirming that the incremental finding is robust to the PS estimator.

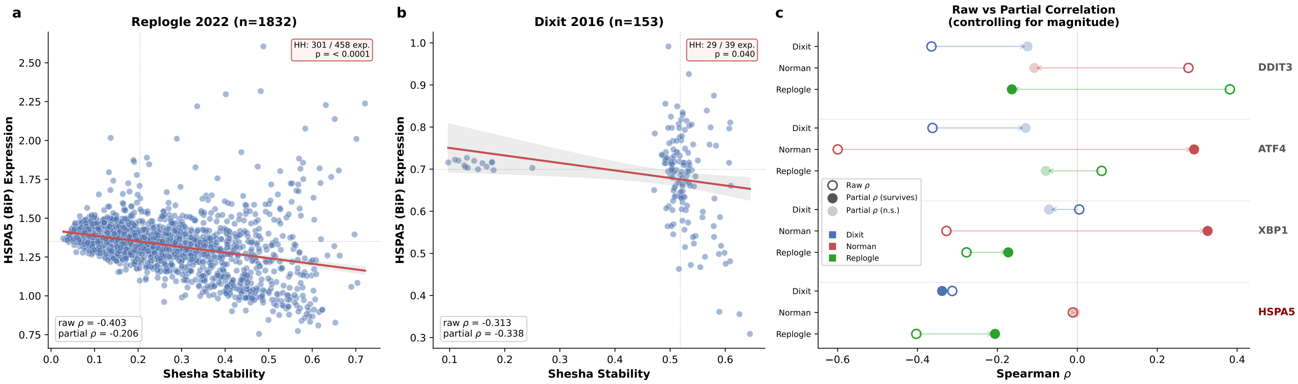

Figure 5: **Geometric instability is independently associated with cellular stress activation.** (**a**) Perturbation stability (x-axis) versus mean HSPA5 expression (y-axis) in the Replogle 2022 CRISPRi dataset (n = 1,832). Red line: linear regression with 95% confidence band. Dotted lines: median splits defining quadrants. The high-stability/high-stress (HH) quadrant is depleted: 301 observed versus 458 expected under independence ($p < 10^{-18}$, one-sided binomial). Raw Spearman $\rho = -0.403$; partial $\rho = -0.206$ after controlling for magnitude. (**b**) Same analysis in Dixit 2016 CRISPRi (n = 153). HSPA5 shows the strongest single-marker partial correlation (partial $\rho = -0.338$, medium-large effect). HH quadrant depleted ($p = 0.040$). (**c**) Raw (open circles) versus partial (filled circles) Spearman correlations between stability and four individual stress markers across three datasets. Filled markers indicate partial correlations surviving magnitude control; open markers show raw correlations. DDIT3 raw correlations collapse or flip sign after partialling out magnitude, while HSPA5 partial correlations persist in both CRISPRi datasets. Pathway-level UPR results, which provide the primary stress finding, are reported in the main text and **SI Appendix, Table S21**.

Geometric instability is associated with cellular stress

The geometric tax describes a structural property of perturbation responses, but does it have functional consequences? We hypothesized that geometrically incoherent perturbations, which scatter cells across expression space rather than guiding them along established trajectories, would be associated with elevated cellular stress. Cells pushed into configurations that do not correspond to stable attractor states may activate homeostatic stress responses as they attempt to restore a viable gene expression program.

Individual stress markers suffer from sign heterogeneity across datasets and modalities (see below), so we first assessed stress at the pathway level. We computed composite pathway scores for four MSigDB Hallmark gene sets (Unfolded Protein Response, p53 Pathway, Apoptosis, Reactive Oxygen Species) using `scanpy.tl.score_genes` with 72–78 overlapping genes for UPR, 32–51 for p53, 53–73 for Apoptosis, and 45–55 for ROS, then correlated these scores with perturbation stability controlling for effect magnitude.

The UPR pathway score showed a sign-consistent negative partial correlation with stability across all four testable datasets (**Fig. 5c**): Replogle (partial $\rho = -0.214$, 95% CI $[-0.273, -0.158]$, $p = 1.7\times10^{-20}$), Dixit (partial $\rho = -0.231$, $[-0.383, -0.064]$, $p = 4.1 \times 10^{-3}$), Papalexi (partial $\rho = -0.395$, $[-0.814, 0.046]$, $p = 0.051$; direction consistent but underpowered at $n = 25$), and Norman (partial $\rho = -0.023$, $[-0.158, 0.117]$, $p = 0.73$). The association survived magnitude control in both large CRISPRi datasets (Dixit and Replogle) after Benjamini-Hochberg correction. No other pathway achieved sign consistency: Apoptosis, p53, and ROS each showed at least one sign reversal across datasets. In the Replogle dataset, functional category stratification confirmed that the UPR-stability association was not driven by any single gene class. The partial correlation among the 1,576 perturbations not assigned to any curated functional category ($\rho = -0.258$, $[-0.324, -0.193]$, $p = 2.2 \times 10^{-25}$) was comparable to the global estimate, indicating a broad association rather than a category-specific confound (**SI Appendix, Table S23**).

Apoptosis provided a secondary stress axis. Its partial correlations with stability were the largest in absolute magnitude among the four pathways in both Norman (partial $\rho = -0.324$, $[-0.463, -0.163]$, $R^2 = 10.5\%$) and Replogle (partial $\rho = -0.333$, $[-0.383, -0.282]$, $R^2 = 11.1\%$), but the association was absent in Dixit (partial $\rho = -0.079$) and reversed in Papalexi (partial $\rho = +0.232$). This cross-dataset inconsistency limits Apoptosis to a dataset-specific observation rather than a generalizable finding, in contrast to the UPR result.

At the individual marker level, HSPA5 (BiP/GRP78), a canonical ER chaperone and UPR activation marker (Oyadomari and Mori 2003), was consistent with the pathway-level result. After controlling for magnitude, the partial correlation between stability and HSPA5 expression was significant in both CRISPRi datasets: Dixit (partial $\rho = -0.338$, $[-0.506, -0.164]$,

$p = 1.9 \times 10^{-5}$; medium-large effect) and Replogle (partial $\rho = -0.206$, $[-0.260, -0.152]$, $p = 5.2 \times 10^{-19}$; small-medium effect) (**Fig. 5a,b**). A quadrant depletion test confirmed this pattern: for HSPA5, the high-stability/high-stress quadrant was depleted in both Dixit ($p = 0.040$, one-sided binomial) and Replogle ($p < 10^{-18}$; 301 observed versus 458 expected, odds ratio = 0.24). The association was null in Norman CRISPRa (partial $\rho = -0.006$). By contrast, DDIT3, a commonly used marker of the integrated stress response, was largely confounded by magnitude: raw correlations were significant in all three datasets but collapsed or reversed sign after partialling out effect size (**Fig. 5c**; **SI Appendix, Robustness Analyses**).

The attenuation of the stress association in Norman CRISPRa was consistent across both pathway-level and single-marker analyses (UPR: partial $\rho = -0.023$; HSPA5: partial $\rho = -0.006$; Apoptosis: partial $\rho = -0.324$ was the sole exception). Gene activation may engage regulatory programs differently from interference: CRISPRa drives cells toward gain-of-function states along existing developmental trajectories, while CRISPRi loss-of-function perturbations are more likely to push cells into off-manifold configurations that trigger homeostatic stress. Notably, the CRISPRa partial correlations are near zero rather than reversed in sign, consistent with the UPR-stability association being attenuated in activation modalities rather than absent or contradicted. A magnitude-stratified analysis of the Norman dataset confirmed the attenuation: three of four magnitude quartiles showed directionally positive HSPA5 elevation in high-discordance perturbations, but none reached significance (Q4: $\Delta = +0.027$, $p = 0.070$; **SI Appendix, Table S22**), consistent with a weak signal diluted by low power ($n = 59$ per bin) rather than a true absence. Whether this reflects a genuine modality dependence or insufficient power in the Norman dataset ($n = 236$ versus 1,832 for Replogle) cannot be resolved without matched CRISPRa/CRISPRi panels targeting the same genes.

Taken together, these results indicate that geometric instability carries functional significance beyond structural description. Perturbations that scatter cells incoherently are independently associated with activation of the unfolded protein response, and this association generalizes across CRISPRi datasets, cell types, and functional gene categories. The effect is modest in absolute terms (4–5% of residual variance for UPR after magnitude control), but it is consistent in direction and robust to the choice of aggregation level (pathway composite versus individual marker). The stress-stability relationship remains correlative: we have not demonstrated that geometric incoherence causes UPR activation, only that the two co-occur after controlling for magnitude.

Stability is a property of biological state space, not linear projection

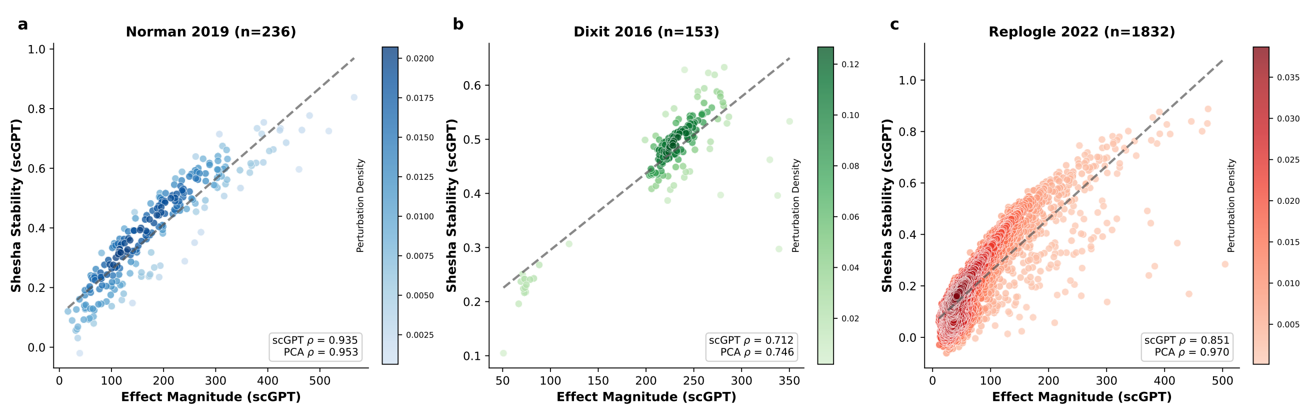

Figure 6: **The magnitude-stability relationship persists in nonlinear foundation model embeddings** Effect magnitude (x-axis) vs perturbation stability (y-axis) computed in scGPT "Whole Human" embeddings for three datasets: (**a**) Norman 2019 ($n = 236$, scGPT $\rho = 0.935$, PCA $\rho = 0.953$), (**b**) Dixit 2016 ($n = 153$, scGPT $\rho = 0.712$, PCA $\rho = 0.746$), (**c**) Replogle 2022 ($n = 1{,}832$, scGPT $\rho = 0.851$, PCA $\rho = 0.970$). Dashed lines: linear regression. The dataset rank order is preserved across embedding methods (Norman > Replogle > Dixit). scGPT correlations are consistently slightly lower than PCA, consistent with the nonlinear embedding resolving additional manifold structure that PCA collapses. The largest drop occurs in Replogle, the most diverse screen.

The results presented thus far rely on PCA embeddings, which project high-dimensional transcriptomic data onto a linear subspace. If the magnitude-stability relationship were an artifact of this linear projection, it would have limited biological significance. To test this, we replaced PCA with scGPT (Cui et al. 2024), a transformer-based foundation model pre-trained on 33 million human cells that learns nonlinear representations of cell state. We computed perturbation stability and magnitude in scGPT embeddings for three datasets (Norman, Dixit, Replogle) using the "Whole Human" pretrained checkpoint, with stability and magnitude computed identically to the PCA pipeline (full protocol in **SI Appendix, scGPT Validation Protocol**).

The magnitude-stability relationship persisted in every dataset (**Fig. 6**). Spearman correlations in scGPT embeddings were $\rho = 0.935$ for Norman (95% CI [0.911, 0.951]), $\rho = 0.712$ for Dixit ([0.585, 0.818]), and $\rho = 0.851$ for Replogle ([0.836, 0.865]), all highly significant ($p < 10^{-25}$). The dataset rank order was preserved across embedding methods (Norman > Replogle > Dixit), confirming that the framework captures real between-dataset variation rather than a ceiling effect.

The scGPT correlations were consistently slightly lower than their PCA counterparts (0.935 vs 0.953 for Norman; 0.712 vs 0.746 for Dixit; 0.851 vs 0.970 for Replogle). This is expected: a nonlinear embedding that resolves manifold structure flattened by PCA will introduce additional geometric complexity. Perturbations that appear coherent in a linear projection may reveal substructure, such as bifurcating trajectories or off-manifold curvature, in the learned embedding. The largest drop occurred in Replogle (0.970 to 0.851), consistent with the genome-scale screen containing the most diverse perturbation types where nonlinear geometry matters most.

These results establish that the magnitude-stability relationship is a property of biological state space rather than an artifact of the embedding method. They also suggest a practical criterion for evaluating foundation model representations: models that preserve the stability-magnitude structure have learned something geometrically faithful about cellular dynamics.

Combinatorial perturbations exhibit higher geometric stability

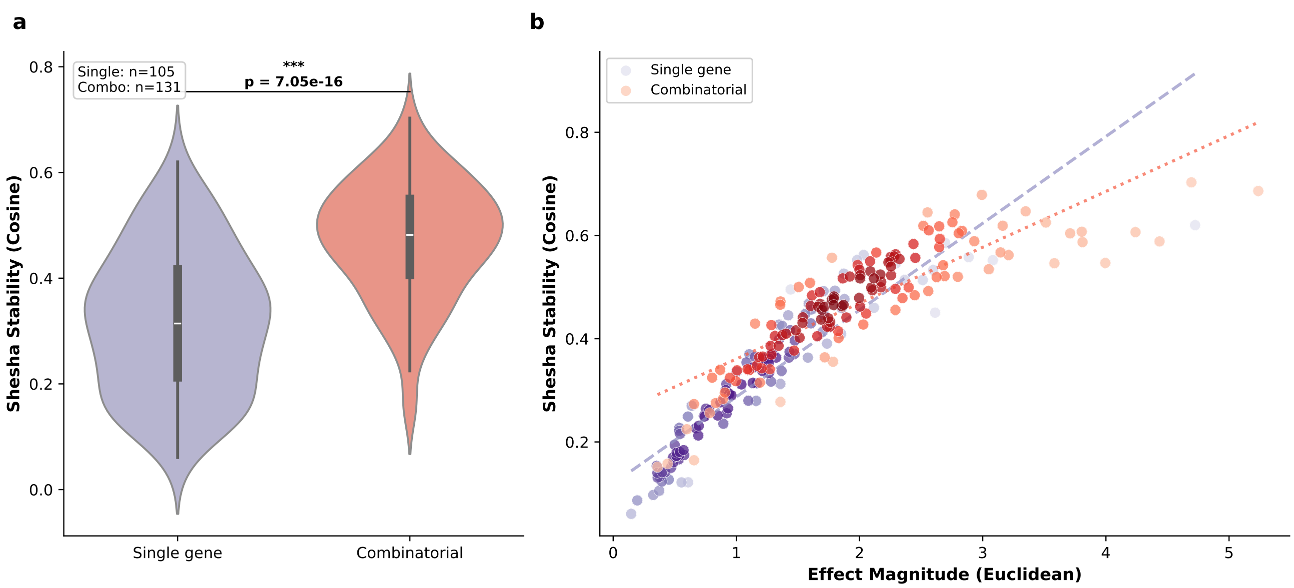

Figure 7: **Combinatorial perturbations exhibit higher geometric stability than single-gene perturbations** (**a**) Distribution of perturbation stability (S_p) for single-gene ($n = 105$) vs combinatorial ($n = 131$) perturbations in the Norman 2019 CRISPRa dataset. Combinatorial perturbations show significantly higher stability (Mann-Whitney U test). Boxes indicate interquartile range; internal line indicates median. (**b**) Magnitude-stability scatter colored by perturbation type. The magnitude-stability relationship holds within both categories with similar regression slopes, indicating that the higher stability of combinatorial perturbations is not explained by their larger effect magnitudes. Dashed line: single-gene regression; dotted line: combinatorial regression.

The Norman CRISPRa dataset contains both single-gene perturbations ($n = 105$) and combinatorial perturbations targeting two genes simultaneously ($n = 131$). Combinatorial perturbations showed significantly higher geometric stability than single-gene perturbations ($p < 10^{-9}$, Mann-Whitney U test; **Fig. 7a**), and this difference was not explained by combinatorial perturbations simply having larger effect magnitudes: the magnitude-stability relationship held within both categories, with similar regression slopes (**Fig. 7b**).

This observation is consistent with the Waddington landscape framework. When two genes are perturbed simultaneously, the combined intervention is more likely to engage an existing developmental trajectory, effectively pushing cells into a deeper attractor basin rather than scattering them off-manifold. Single-gene perturbations, by contrast, may activate only one component of a multi-factor regulatory program, producing a partial shift that lacks the coordinated directionality of a full lineage transition. The higher coherence of combinatorial perturbations suggests that they more frequently align with canalized developmental programs, where the epigenetic landscape itself channels cells toward a stable endpoint.

Discussion

The geometric tax: a framework for interpreting perturbation coherence

The empirical results point to a consistent pattern. Perturbation magnitude and geometric stability are strongly correlated across datasets, modalities, and embedding methods, but they decouple in a biologically structured way: perturbations targeting pleiotropic regulators produce large but incoherent responses, while perturbations targeting lineage-specific factors produce coherent responses that align with established developmental trajectories. We propose the term "geometric tax" to describe this cost, measured in directional coherence, of activating broad regulatory programs.

The geometric tax is a consequence of regulatory network topology. When a transcription factor such as CEBPA (Friedman 2007) engages dozens of competing downstream pathways, no single cell can activate all target programs simultaneously. Each cell resolves the competition differently, producing a population-level incoherence that is invisible to standard effect-size metrics. By contrast, lineage-specific factors such as KLF1 (Siatecka and James J. Bieker 2011; Tallack, Whitington, et al. 2010), whose targets are functionally coordinated toward a single developmental outcome, maintain geometric coherence because the downstream programs are locally collinear on the state manifold. The LOESS-corrected discordance analysis in Replogle extends this principle beyond transcription factors: the most discordant perturbations are subunits of multi-protein complexes (spliceosome components SF3B2, SF3B3; ESCRT factors CHMP2A, CHMP3; proteasome subunit PSMD7). Loss of one subunit in a large molecular machine produces heterogeneous downstream consequences because the residual complex activity, the rate-limiting processing step, and the cell's demand for that machinery all vary from cell to cell. The geometric tax thus applies to any perturbation whose downstream targets span multiple functional programs, whether those targets are transcriptional (CEBPA) or post-transcriptional (spliceosome, ESCRT).

The functional diversity analysis confirms this dual-mechanism interpretation. CEBP-family perturbations engage more than twice as many GO Biological Process categories as KLF1 combinations ($p = 0.013$), consistent with their geometric incoherence arising from broad transcriptional scope. Yet functional diversity does not explain discordance globally ($\rho = -0.119$, $p = 0.11$), which is expected given that the Replogle top-discordant genes are spliceosome and ESCRT subunits whose incoherence stems from heterogeneous partial loss-of-function rather than from diverse transcriptional programs. The geometric tax thus has at least two mechanistic sources: target diversity for pleiotropic transcription factors, and stochastic complex disassembly for multi-subunit molecular machines.

This framework connects to biological robustness more broadly. Gene regulatory networks are optimized for the stability of expression patterns under perturbation, not just for the patterns themselves (Kitano 2004; Siegal and Bergman 2002). The geometric tax quantifies the cost of working against this optimization. In the language of Waddington's epigenetic landscape, perturbations into deep valleys, where canalized programs channel cells toward robust attractors, pay minimal tax (Davila-Velderrain, Martinez-Garcia, and Alvarez-Buylla 2015). Perturbations onto ridges or flat regions, where no strong attractor constrains the trajectory, pay heavily.

This interpretation aligns with recent experimental evidence from Wilson et al. (Wilson et al. 2025), who systematically disrupted 200 epigenetic regulator genes and found that network robustness emerges from functional cooperation and degeneracy among components rather than from the resilience of individual genes. In their framework, perturbations of genes with redundant functional partners are buffered by the network, while perturbations of non-redundant components expose synthetic fragilities. The geometric tax provides a quantitative readout of this distinction: buffered perturbations should produce coherent cellular responses (high S_p, cells channeled toward the same compensated state), while perturbations that expose network fragilities should scatter cells across expression space as each cell resolves the loss differently.

Relationship to existing methods

Several recent methods address within-perturbation heterogeneity from complementary angles. Nadig et al. (Nadig et al. 2025) introduced TRADE, a statistical framework that models the distribution of true differential expression effects across genes, accounting for estimation error in pseudobulk log-fold-change estimates. TRADE's primary output, transcriptome-wide impact (TI = Var(β)), quantifies the total perturbation effect across the transcriptome, while the effective number of differentially expressed genes (π_{DEG}) captures whether that effect is concentrated in a few targets or spread across hundreds. S_p addresses a question that TRADE does not: among the cells that were affected, did they change in the same direction? TRADE operates on pseudobulk summaries and characterizes the gene-level breadth of a perturbation's effect; S_p operates on individual cell shift vectors and characterizes the population-level geometric coherence of the response. A perturbation could have high TI and large π_{DEG} (broad transcriptomic changes across many genes) yet low S_p (different cells implementing different subsets of those changes), or moderate TI with high S_p (a focused effect applied uniformly across the population). Nadig et al.'s finding that essential gene perturbations affect over 500 genes on average is consistent with our observation that essential genes frequently pay the geometric tax: broad downstream effects are more likely to fragment across cell subpopulations, producing geometric incoherence. A direct comparison of TI and π_{DEG} with S_p across matched perturbations would clarify the relationship between transcriptomic breadth and geometric coherence.

A preliminary comparison using a TRADE-style variance decomposition (eta^2, the proportion of PCA variance explained by perturbation identity) reveals that Sp-discordance and eta^2 are negatively correlated in Replogle ($\rho = -0.276$, $p < 10^{-33}$): perturbations that scatter cells geometrically explain less transcriptomic variance in aggregate, consistent with geometric incoherence diluting the population-level signal that pseudobulk methods depend on (**SI Appendix, Table S15**).

The perturbation-response score (PS) of Song et al. (Song et al. 2025) quantifies how strongly each individual cell responded to a perturbation. A three-tier comparison (centroid-based Euclidean and Mahalanobis proxies, plus a Python port of Song et al.'s scMAGeCK algorithm) revealed that centroid-based distance does not approximate the actual PS ($\rho = 0.097$); centroid-based proxies anticorrelate with S_p after magnitude control, but this reflects the mechanical relationship between coherence and per-cell scatter rather than a property of Song et al.'s score. With the algorithm-derived PS, S_p and PS are moderately positively correlated after controlling for magnitude (partial $\rho = +0.507$), sharing approximately 20% of variance. S_p provides significant incremental prediction of UPR pathway activation beyond both magnitude and the real PS ($\rho = -0.203$, $p < 10^{-18}$), while the real PS adds only weakly beyond S_p ($\rho = -0.072$, $p = 0.002$). This asymmetry indicates that the directional coherence captured by S_p contains stress-relevant information that per-cell response strength does not fully account for.

Functional consequences of geometric instability

The association between geometric instability and activation of the unfolded protein response provides functional evidence that the geometric tax carries biological costs. Cells scattered into off-manifold configurations that do not correspond to stable attractor states activate the UPR (Lee 2005; Ron and Walter 2007), suggesting that geometric incoherence incurs a measurable homeostatic cost (Hetz 2012; Walter and Ron 2011). That this association is sign-consistent across all four testable datasets at the pathway level, and that it survives after controlling for effect magnitude, indicates that the stress response is linked to the geometry of the perturbation rather than its strength alone. The effect is modest (4–5% of residual variance for UPR), but its consistency across datasets, cell types, and aggregation levels (pathway composite versus individual marker) argues against a statistical artifact. The stress-stability relationship remains correlative: establishing causality would require experimental manipulation of perturbation coherence independent of magnitude, for example through titrated guide RNA delivery or inducible systems that control the rate of target gene knockdown.

Practical applications

The split-half reproducibility analysis provides direct evidence for one practical application: hit prioritization in CRISPR screens. At equivalent magnitude, perturbations with higher S_p are more likely to produce directionally reproducible phenotypes, and this advantage is consistent across all magnitude strata. When faced with a list of screen hits of similar effect size, S_p provides a principled secondary ranking criterion that predicts which effects will replicate.

Two additional applications warrant exploration but lack direct experimental support in this study. In cell manufacturing, geometric stability could complement marker-based quality control (Bravery et al. 2013; Galipeau et al. 2016): a cell product with low S_p may occupy a flat region of the manifold where small perturbations could redirect trajectories, even if the product passes conventional marker-based criteria. In regulatory evaluation, incorporating geometric stability alongside existing potency and identity assays (Salmikangas et al. 2023; Simon et al. 2024) could address failure modes that current frameworks do not access, such as geometric drift toward unintended fates in CAR-T (Fraietta et al. 2018; Philip et al. 2017) or iPSC-derived cell products (Lipsitz, Timmins, and Zandstra 2016; Nair et al. 2019). These applications remain speculative and would require prospective validation in manufacturing and clinical contexts.

Foundation models and in silico perturbation prediction

The magnitude-stability relationship persists in scGPT embeddings, which extends the finding beyond methodological robustness. Foundation models for single-cell biology, including scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), and UCE (Rosen et al. 2023), learn implicit representations of cell state geometry. Preservation of the stability-magnitude structure provides a necessary, though not sufficient, criterion for evaluating whether these learned representations capture biologically meaningful geometry.

For in silico perturbation prediction tools such as GEARS (Roohani, K. Huang, and Leskovec 2023), CellOracle (Kamimoto et al. 2023), and CPA (Lotfollahi et al. 2023), geometric stability offers a complementary evaluation metric. A predicted cell state with high magnitude shift but low predicted stability may represent a computationally plausible but biologically unstable configuration that no real cell would occupy for long. Conversely, predictions that maintain high coherence are more likely to correspond to viable attractor states.

Next-generation Perturb-seq datasets will provide natural testing grounds for geometric stability analysis. Spatially resolved screens such as PerturbSpace (Nevue et al. 2026), which integrates CRISPR perturbations with spatial transcriptomics in vivo, would enable testing whether geometric incoherence in dissociated screens reflects cell-intrinsic regulatory properties or is modulated by tissue microenvironment. Genome-scale screens in primary human cells such as the CD4$^+$ T cell atlas of Zhu et al. (Zhu et al. 2025), which profiles perturbation effects across resting and stimulated conditions, would test whether the geometric tax generalizes beyond immortalized cell lines to primary cells with context-dependent regulatory networks.

Combinatorial perturbations

Combinatorial perturbations in the Norman dataset showed higher geometric stability than single-gene perturbations, even after controlling for their larger effect magnitudes. This is consistent with the hypothesis that simultaneously engaging two components of a regulatory program is more likely to align with a canalized developmental trajectory than engaging one component alone. However, as Roohani et al. (Roohani, K. Huang, and Leskovec 2023) demonstrated with GEARS, many combinatorial perturbation effects are near-additive, and Norman et al. selected gene pairs expected to interact. The higher stability of combinatorial perturbations in this dataset may therefore partly reflect selection for interacting pairs rather than a general property of multi-gene perturbations. Testing this hypothesis would require combinatorial screens with randomly paired genes.

Limitations

Several limitations should guide the interpretation of these results.

PCA serves as the primary embedding throughout, though the scGPT validation mitigates concerns regarding linear projection artifacts. Manifold-aware methods such as diffusion maps or PHATE (Moon et al. 2019) may reveal additional nonlinear structure.

The Adamson dataset (Adamson et al. 2016) ($n = 8$) provides limited statistical power, as reflected in its wide bootstrap confidence intervals. The perturbations available through pertpy for this dataset are transcription factor knockdowns rather than the UPR-targeting arm of the original Adamson et al. study, precluding its use as a positive control for the stress-stability association.

Discordance rankings are sensitive to the choice of residual method. We report LOESS-based residuals in the main text; linear and rank-based alternatives are provided in SI Appendix. The top discordant genes differ substantially between methods for high-magnitude perturbations where the nonlinearity in the magnitude-stability relationship is strongest.

Biological conclusions about which specific genes are most discordant should be interpreted with this sensitivity in mind.

The stress-stability association remains correlative rather than causal. The pathway-level UPR analysis improved on the original single-marker approach by resolving sign heterogeneity across datasets, but the effect sizes remain modest (4–5% of residual variance). We have not demonstrated that geometric incoherence causes UPR activation, only that the two co-occur after controlling for magnitude.

S_p operates as a global metric that summarizes each perturbation as a single scalar; it does not capture subpopulation-level structure, bifurcating responses, or dose-dependent heterogeneity within a single perturbation. Our analysis operates at the gene level rather than the guide level, meaning that unmeasured guide-level variation in perturbation efficiency could contribute to apparent incoherence, particularly for high-magnitude perturbations where multiple guides with different efficiencies may be pooled.

Our initial comparison of S_p against Song et al.'s perturbation-response score used centroid-based distance proxies that proved to be poor approximations of the actual PS ($\rho < 0.15$). Results reported using the algorithm-derived PS supersede the proxy-based analyses. Both the incremental UPR prediction and the split-half reproducibility advantage are robust across all three PS tiers (Euclidean proxy, Mahalanobis proxy, and scMAGeCK algorithm), but the relationship between S_p and PS (positive correlation rather than anticorrelation) differs qualitatively from the proxy-based result, underscoring the importance of using authors' implementations rather than simplified approximations when comparing metrics. To facilitate accurate reproduction of our own metric, S_p is implemented in the open-source `shesha-geometry` package (PyPI), requiring only an AnnData object with perturbation labels and a control condition.

Despite these limitations, the consistency of the magnitude-stability relationship across five datasets, two perturbation modalities, multiple cell types, and two fundamentally different embedding methods suggests that geometric stability captures a robust property of how cells respond to genetic perturbation. The geometric tax framework provides a complementary axis for evaluating perturbation screens, assessing screen hit reproducibility, and characterizing the regulatory architecture that shapes cellular responses to intervention.

Acknowledgments

We thank Padma K. and Annapoorna Raju for generously supporting the computational resources used in this work. We thank the many institutions and individuals whose open-source datasets, frameworks, and models were used in our work. The authors acknowledge the use of large language models (specifically the GPT, Claude, and Gemini families) to assist with code debugging and text polishing. All hypotheses, experimental designs, analyses, and interpretations were independently formulated and verified by the authors, and the authors assume full responsibility for all content and claims in this work.

Code and data availability

All analyses reported in this paper, including figure generation code, are available at https://github.com/prashantcraju/geometric-stability-crispr. All five datasets are publicly available through pertpy Heumos et al. 2025 and require no login or registration. The Shesha perturbation stability metric is implemented in the open-source Python package `shesha-geometry`, freely available on PyPI (https://pypi.org/project/shesha-geometry; Raju 2026b) under an MIT license with accompanying tutorials.

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Extended Methods

Datasets and preprocessing

Five single-cell CRISPR perturbation datasets were accessed via the pertpy Python package (version 1.0.4) (Heumos et al. 2025). Table S1 summarizes the datasets. Each dataset was preprocessed independently to prevent batch effects. The pipeline for each dataset consisted of:

1. Quality filtering: cells with fewer than 100 detected genes were removed.
2. Library-size normalization (Wolf, Angerer, and Theis 2018): `scanpy.pp.normalize_total()` with default parameters (Norman, Dixit, Papalexi) or `target_sum=1e4` (Replogle, Adamson).
3. Log transformation: `scanpy.pp.log1p()`.
4. Highly variable gene selection: `scanpy.pp.highly_variable_genes(n_top_genes=2000, subset=True)`.
5. PCA: `scanpy.tl.pca(n_comps=50)`.

All downstream stability and magnitude computations were performed on the 50-dimensional PCA embedding.

Control group identification

Control group assignment used a multi-stage matching protocol to accommodate heterogeneous labeling conventions across datasets:

1. **Exact match** (case-insensitive): labels matching "control", "ctrl", "non-targeting", "NT", "unperturbed".
2. **Delimiter-aware regex**: for short tokens (e.g., "NT" in "NTg1"), split on common delimiters and match components.
3. **Substring matching**: for longer keywords embedded in complex labels.

Dataset-specific handling:

- **Replogle 2022:** Labels containing "non-targeting" or beginning with "chr" were assigned to control. Labels containing "pos_control" were removed. Gene names were extracted by splitting on underscore and taking the first token.
- **Papalexi 2021:** The `gene_target` column was copied from MuData global metadata. Non-targeting guides (NTg1 through NTg7) were pooled into a single NT control group (2,386 cells). Note: the pertpy loader for Papalexi is incompatible with current mudata versions due to hashed MuData keys; results for this dataset are from a prior pipeline run with identical preprocessing parameters.
- **Norman 2019:** The `perturbation_name` column was used directly. Cells labeled "control" served as the control group.

Perturbation stability and effect magnitude

For a perturbation p applied to n_p cells, let $\mathbf{x}_i \in \mathbb{R}^{50}$ denote the PCA coordinates of perturbed cell i and $\boldsymbol{\mu}_{\text{ctrl}}$ the centroid of all control cells in the same dataset. The shift vector for cell i is:

$$\mathbf{d}_i = \mathbf{x}_i - \boldsymbol{\mu}_{\text{ctrl}} \tag{2}$$

The mean perturbation direction is:

$$\bar{\mathbf{d}} = \frac{1}{n_p} \sum_{i=1}^{n_p} \mathbf{d}_i \tag{3}$$

Shesha perturbation stability (S_p) is defined as:

$$S_p = \frac{1}{n_p} \sum_{i=1}^{n_p} \frac{\mathbf{d}_i \cdot \bar{\mathbf{d}}}{\|\mathbf{d}_i\| \, \|\bar{\mathbf{d}}\|} \tag{4}$$

Effect magnitude is:

$$M_p = \|\bar{\mathbf{d}}\| \tag{5}$$

Discordance is the standardized residual from the ordinary least-squares regression of S_p on M_p:

$$D_p = z(M_p) - z(S_p) \tag{6}$$

where $z(\cdot)$ denotes within-dataset z-score normalization. Positive discordance indicates that a perturbation has lower stability than predicted by its magnitude (below the regression line); negative discordance indicates higher stability than predicted (above the line).

The perturbation stability metric adapts the principle of geometric self-consistency from the Shesha representational stability framework (Raju 2026a,b). In the general framework, Shesha-FS measures split-half RDM consistency across representations. Here, S_p measures perturbation coherence directly within a single representation, specializing the principle to perturbation biology.

Minimum cell count filtering

Perturbations with fewer than 50 cells (10 for Dixit) were excluded to ensure stable estimates. The Adamson dataset retains all 8 perturbations despite wide bootstrap confidence intervals, which honestly reflect the limited statistical power.

Robustness Analyses

Distance metric robustness

All methods produce comparable or stronger correlations relative to the standard Euclidean metric (Table S4), confirming that the magnitude-stability relationship is robust to the choice of distance metric. Whitened (Mahalanobis-scaled) and k-NN matched methods produce equal or higher correlations across all four datasets, consistent with these methods reducing noise from batch effects and control group heterogeneity.

PCA dimensionality ablation

Stability was recomputed at PCA dimensionalities of 10, 20, 30, 50, and 100 components. Spearman ρ increased monotonically with dimensionality: Norman $\rho = 0.949$ (10 PCs) to 0.969 (100 PCs); Dixit $\rho = 0.781$ (10 PCs) to 0.869 (100 PCs). The default of 50 components ($\rho = 0.959$ for Norman, 0.844 for Dixit) lies in the upper portion of the range, and the relationship is robust across all tested dimensionalities.

Seed reproducibility

Cross-seed Spearman correlation of perturbation-level stability rankings across 15 random seeds (320, 1991, 9, 7258, 7, 2222, 724, 3, 12, 108, 18, 11, 1754, 411, 103) was near-perfect: Norman mean $r = 0.99997$ (range $[0.9999, 1.0000]$, 105 pairwise comparisons); Dixit mean $r = 0.99963$ (range $[0.9993, 0.9999]$). The magnitude-stability Spearman ρ varied by less than 0.001 across seeds for both datasets (Norman: 0.959–0.960; Dixit: 0.844–0.844). All reported results use seed 320.

Leave-one-out influence

The maximum $|\Delta\rho|$ upon removing any single perturbation was 0.0019 for Norman (most influential: HES7; LOO ρ range $[0.959, 0.961]$) and 0.0106 for Dixit (most influential: INTERGENIC1144056+INTERGENIC1216445; LOO ρ range $[0.840, 0.854]$), indicating that no individual perturbation drives the overall correlation.

Theoretical null model

Under a null model where individual cell shift vectors are drawn uniformly on the unit hypersphere (50 dimensions), the expected stability is $S_p \approx 0$ with variance inversely proportional to n_p. Observed stability values (0.05–0.85) far exceed this null, confirming that the coherence signal is biological rather than statistical.

Mixed-Effects Model

To assess cross-dataset generalization and identify confounds, we fit a linear mixed-effects model:

$$S_p = \beta_0 + \beta_1 \cdot M_p + \beta_2 \cdot \text{spread}_p + \beta_3 \cdot n_p + u_{\text{dataset}} + \varepsilon_p \tag{7}$$

where M_p is effect magnitude, spread_p is within-perturbation expression variance, n_p is cell count, u_{dataset} is a dataset-level random intercept, and ε_p is residual error. All predictors were z-scored within each dataset prior to fitting.

Magnitude accounts for approximately 11 times more variance in stability than sample size ($|\beta_1/\beta_3| = 11.2$), confirming that the magnitude-stability relationship is not driven by differential cell sampling. The near-zero dataset random-effect variance confirms that the relationship generalizes across datasets rather than being driven by any single screen. Full results in Table S2.

Extended Stress Marker Analysis

Partial correlations

Table S14 reports raw and partial Spearman correlations between perturbation stability and four stress/UPR markers across three datasets, controlling for effect magnitude.

Quadrant depletion tests

Perturbations were split at median stability and median stress marker expression. The high-stability/high-stress (HH) quadrant count was compared to expectation under independence using a one-sided binomial test. Full quadrant counts for all four markers across all three datasets are provided in Table S20.

Modality analysis

The DDIT3 sign heterogeneity (positive raw ρ in Norman and Replogle, negative in Dixit) partly reflects CRISPRa vs CRISPRi biology. However, within CRISPRi alone, the sign also differs between Dixit (BMDCs, raw $\rho = -0.365$) and Replogle (K562, raw $\rho = +0.382$), indicating that cell type and experimental design contribute to the heterogeneity. After magnitude control, all three DDIT3 partial correlations are negative, but only Replogle survives with a small effect.

HSPA5 is the most consistent marker across modalities: negative partial correlations in both CRISPRi datasets (Dixit and Replogle), null in CRISPRa (Norman). Full modality-stratified results are provided in Tables S14.

Extended Replogle Analysis

The Replogle 2022 genome-scale CRISPRi screen provides independent validation at unprecedented scale ($n = 1{,}832$ perturbations after filtering for ≥ 50 cells; 310,385 total cells). The Replogle discordance scatter (*SI Appendix*, Fig. S1) shows the same pattern observed in Norman: pleiotropic regulators (GATA1, CHMP3, AQR) cluster below the regression line, while narrowly-acting factors (LSG1, ISG20L2, KRI1) cluster above. See Table S9.

scGPT Validation Protocol

Model and embedding

Cell embeddings were generated using the scGPT “Whole Human” pretrained checkpoint (`scGPT_human`), downloaded from the official repository (https://github.com/bowang-lab/scGPT). The checkpoint contains three files: `best_model.pt`, `vocab.json`, and `args.json`.

Input data consisted of raw counts (the `counts` layer was extracted from each AnnData object; data were not log-normalized prior to embedding). Embeddings were generated using `embed_data()` with `gene_col="index"`, `batch_size=64`, and `use_fast_transformer=False`.

Reproducibility settings

All scGPT computations used deterministic mode:

- Python random seed: 320
- NumPy random seed: 320
- PyTorch manual seed: 320 (CPU and CUDA)
- `torch.backends.cudnn.deterministic = True`
- `torch.backends.cudnn.benchmark = False`

Stability and magnitude computation

Stability and magnitude were computed from scGPT embeddings using `shesha.bio.compute_stability()` and `shesha.bio.compute_magnitude()` with `perturbation_key='perturbation_name'`, `control_label='control'`, and `metric='cosine'` (stability) or `metric='euclidean'` (magnitude). Bootstrap confidence intervals: 10,000 resamples, seed 320, percentile method.

Datasets

scGPT (Cui et al. 2024) analysis was performed on Norman 2019, Dixit 2016, and Replogle 2022. Adamson 2016 ($n = 8$) was included in the output CSV but omitted from main-text figures due to limited power. Papalexi 2021 was excluded due to pertpy loader incompatibility with current mudata versions.

Results

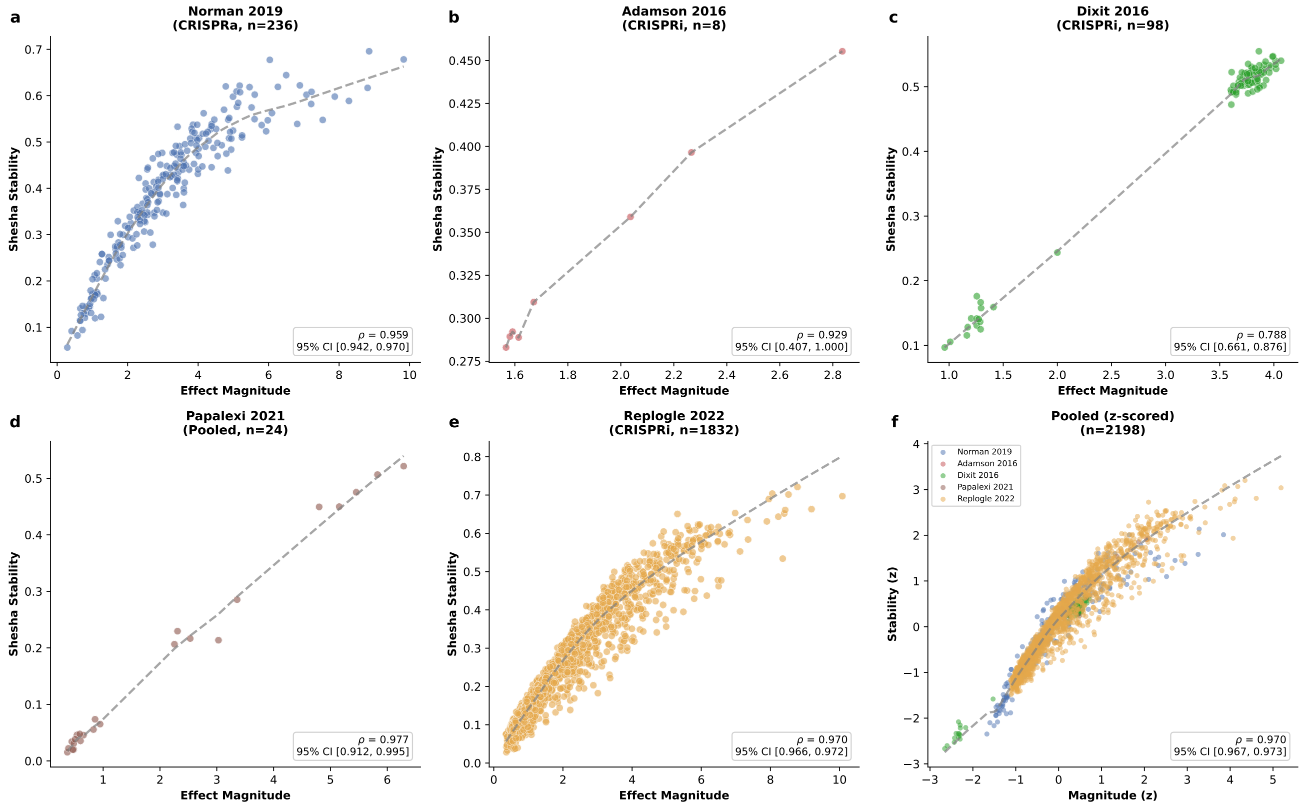
The dataset rank order is preserved (Norman > Replogle > Dixit). scGPT correlations are consistently slightly lower, with the largest drop in Replogle ($\Delta\rho = -0.119$), consistent with the nonlinear embedding resolving manifold structure that PCA collapses. See Table S19

Combinatorial Analysis

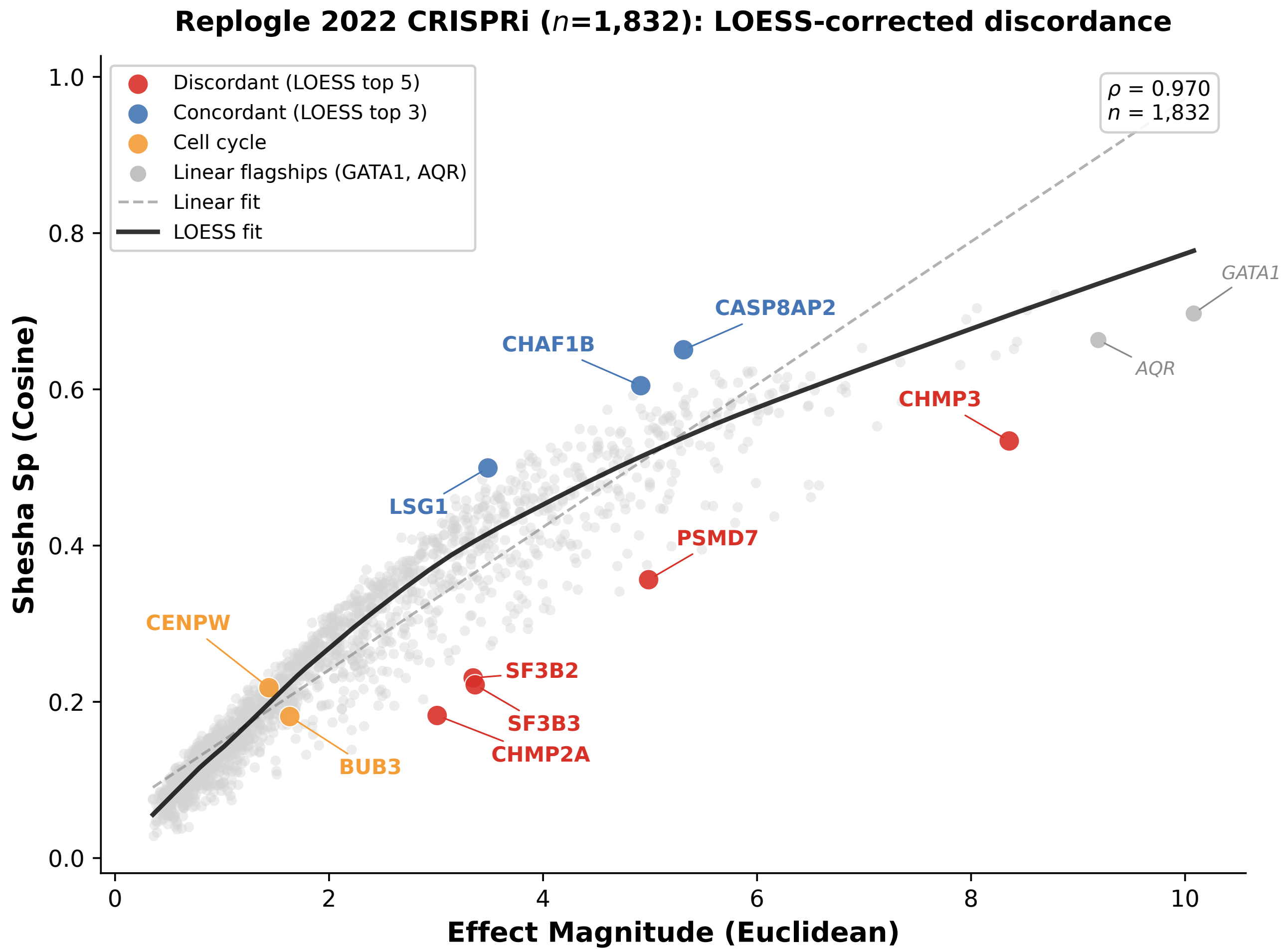
The Norman 2019 dataset contains $n = 105$ single-gene and $n = 131$ combinatorial (two-gene) perturbations. Combinatorial perturbations showed significantly higher stability (mean $S_p = 0.460$ vs 0.306; Mann-Whitney $U = 2{,}637$, $p = 4.1 \times 10^{-16}$).

The magnitude-stability relationship held within both categories, with regression slopes of 0.089 (single-gene, $R = 0.925$, $\rho = 0.973$) and 0.058 (combinatorial, $R = 0.874$, $\rho = 0.919$), confirming that the higher stability of combinatorial perturbations is not simply a consequence of their larger effect magnitudes.

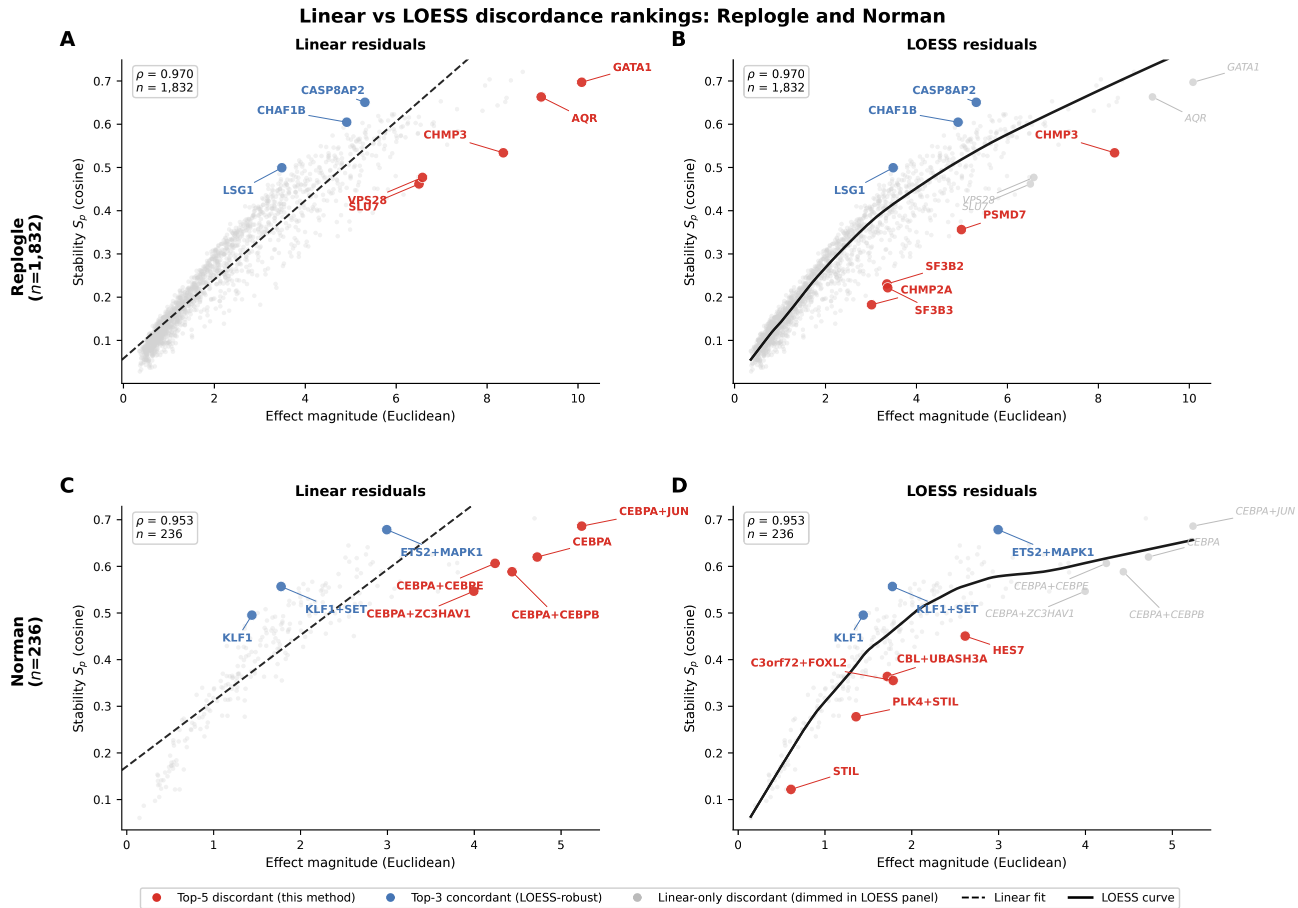
Within the combinatorial set, perturbations involving lineage-specific factors (e.g., KLF1+SET, KLF1+TGFBR2) showed higher stability than those involving pleiotropic factors (e.g., CEBPA+JUN, CEBPA+CEBPB), consistent with the discordance pattern observed in the single-gene analysis.



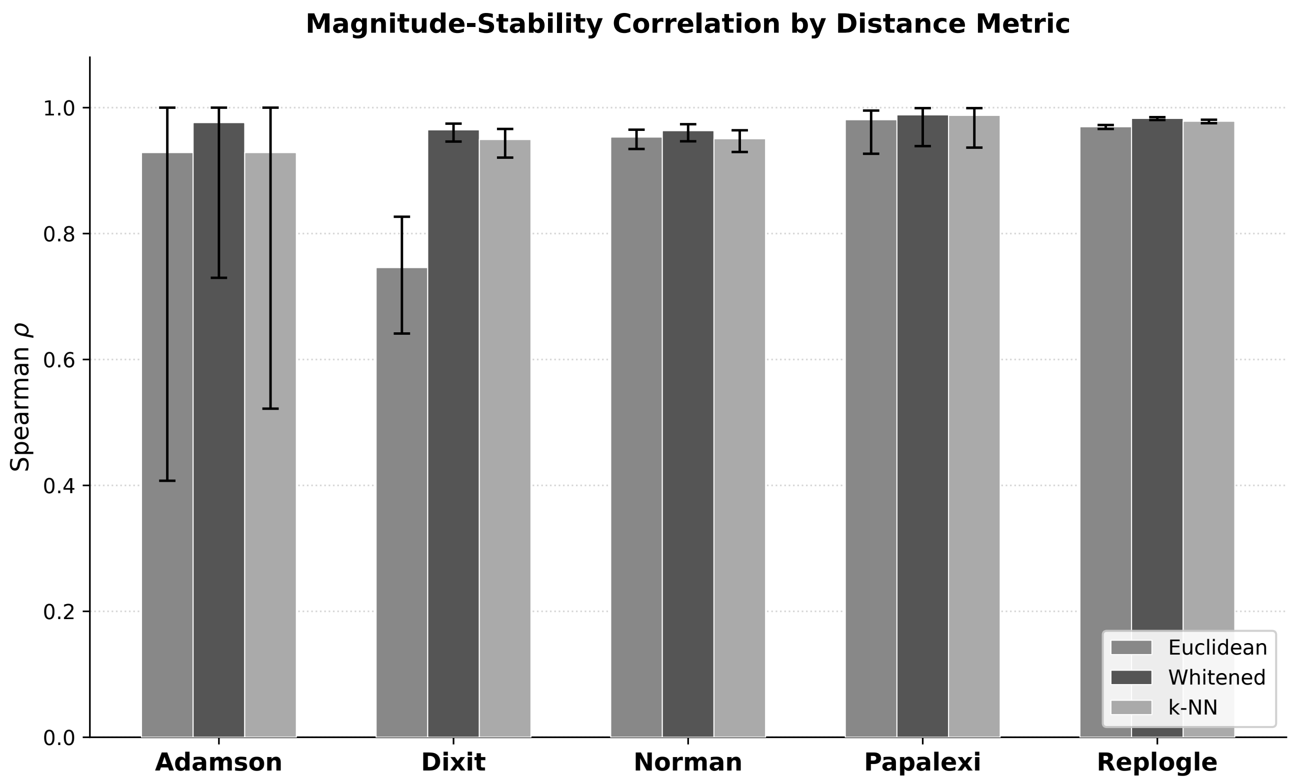
Supplementary Figure S1: **Magnitude-stability relationship with linear and LOESS fits across five CRISPR datasets.** Each panel shows effect magnitude (x-axis) versus Shesha perturbation stability (y-axis) for one dataset. Dashed gray line: linear regression. Solid black curve: LOESS fit (bandwidth fraction = 0.3). Spearman ρ and sample size annotated per panel. The nonlinearity is most pronounced at low magnitudes, where signal-to-noise constrains coherence regardless of the perturbation's biological properties. At high magnitudes (visible in Replogle and Norman), the LOESS curve plateaus while the linear fit continues to rise, explaining why LOESS-corrected discordance rankings differ from linear rankings for high-magnitude perturbations (Fig. S3). Adamson ($n = 8$) is shown for completeness but the LOESS fit is unreliable at this sample size.



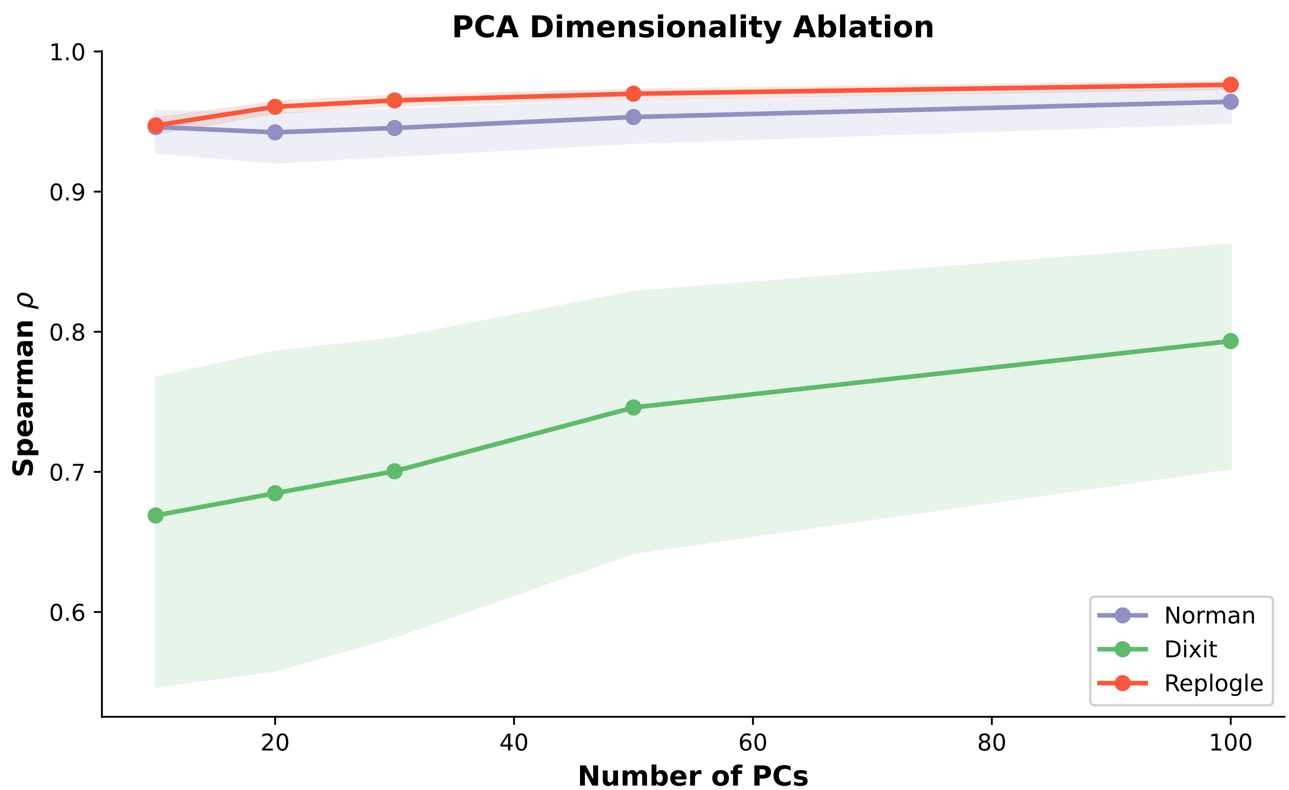
Supplementary Figure S2: **LOESS-corrected discordance analysis in the Replogle 2022 genome-scale CRISPRi screen.** Effect magnitude (Euclidean, x-axis) versus Shesha stability (cosine, y-axis) for n = 1,832 perturbations in Replogle et al. (2022) K562 cells. Solid black line: LOESS fit (bandwidth fraction = 0.3); dashed gray line: linear fit. Points are categorized by LOESS-corrected discordance: *Discordant* (red): low stability relative to the LOESS curve, dominated by subunits of multi-protein complexes: CHMP2A (ESCRT-III, LOESS rank 1), SF3B3 and SF3B2 (U2 snRNP spliceosome, ranks 2–3), PSMD7 (26S proteasome, rank 4), and CHMP3 (ESCRT-III, rank 5). *Concordant* (blue): high stability relative to the LOESS curve, including CASP8AP2 (histone mRNA processing), CHAF1B (chromatin assembly), and LSG1 (ribosome biogenesis), genes with narrow functional scope. *Cell cycle* (orange): BUB3 (spindle assembly checkpoint) and CENPW (centromere protein) show low stability independent of cell cycle arrest. *Linear flagships* (gray, italic): GATA1 and AQR, which ranked 1st and 3rd under linear residuals, sit near the LOESS curve at high magnitudes, indicating that their apparent discordance was driven by the nonlinearity of the magnitude-stability relationship rather than by biology. CHMP3 is the only gene from the original linear top 3 that retains a top-5 ranking under LOESS correction (Table S9).



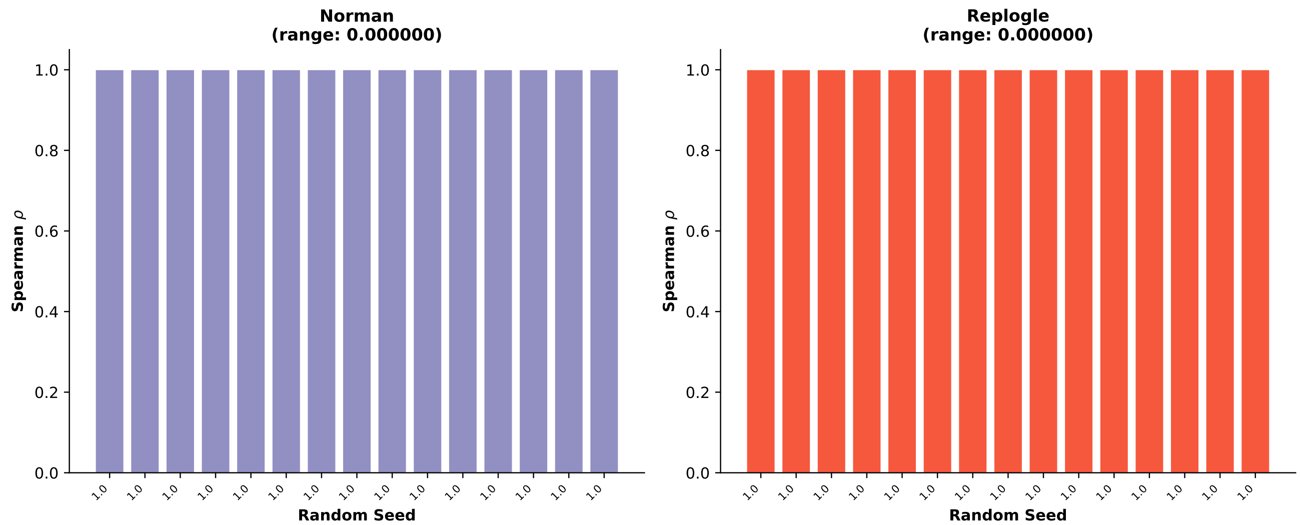
Supplementary Figure S3: **Discordance rankings are sensitive to residual method for high-magnitude perturbations.** (**A, B**) Replogle 2022 CRISPRi (n = 1,832) with linear (A) and LOESS (B) residuals. Under linear residuals, the top discordant genes include GATA1 and AQR at extreme magnitudes. Under LOESS correction, these sit near the fitted curve (gray, italic) and are replaced by spliceosome (SF3B2, SF3B3) and ESCRT (CHMP2A) subunits at moderate magnitudes. CHMP3 retains a top-5 ranking under both methods. Concordant genes (blue) are stable across methods. (**C, D**) Norman 2019 CRISPRa (n = 236) with linear (C) and LOESS (D) residuals. Under linear residuals, CEBPA and its combinations dominate the discordant set. Under LOESS correction, these migrate to near or above the curve (gray, italic), and the top discordant perturbations shift to PLK4+STIL, HES7, and STIL. KLF1 remains the most concordant single-gene perturbation under both methods (linear rank 236, LOESS rank 235), confirming that the concordance finding is method-robust. The sensitivity to residual method is concentrated at high magnitudes where the magnitude-stability relationship is most nonlinear. Biological conclusions about specific gene rankings should be interpreted in light of both methods (Table S7 & S9).



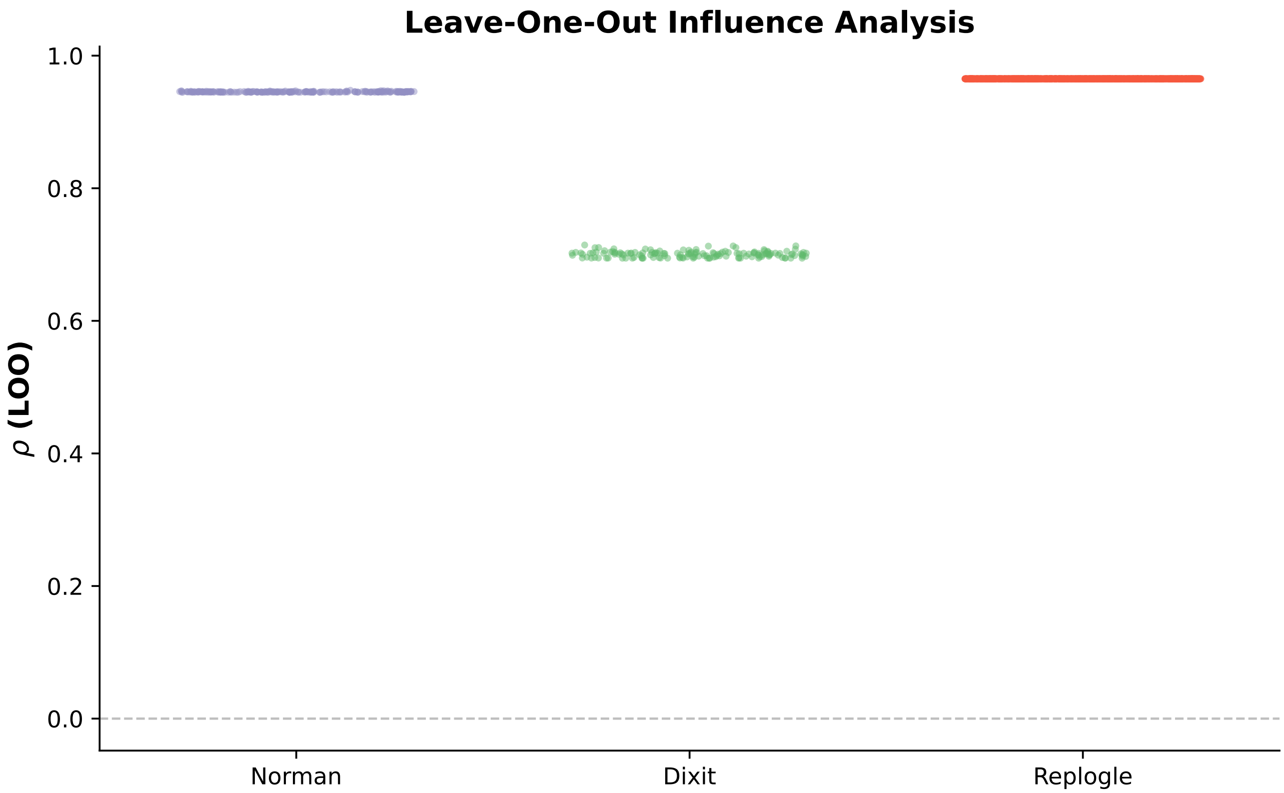
Supplementary Figure S4: **Magnitude-stability correlation is robust across distance metrics.** Bar chart showing Spearman correlations with 95% bootstrap CIs (error bars) for three distance computation methods: Euclidean (standard L_2 in PCA space), Whitened (Mahalanobis-scaled coordinates), and k-NN (local control centroids). All methods achieve strong correlations ($\rho > 0.67$) across all datasets. Whitening substantially improves the Dixit correlation from $\rho = 0.75$ to $\rho = 0.97$, suggesting residual covariance structure in PCA space attenuates the relationship in that dataset.



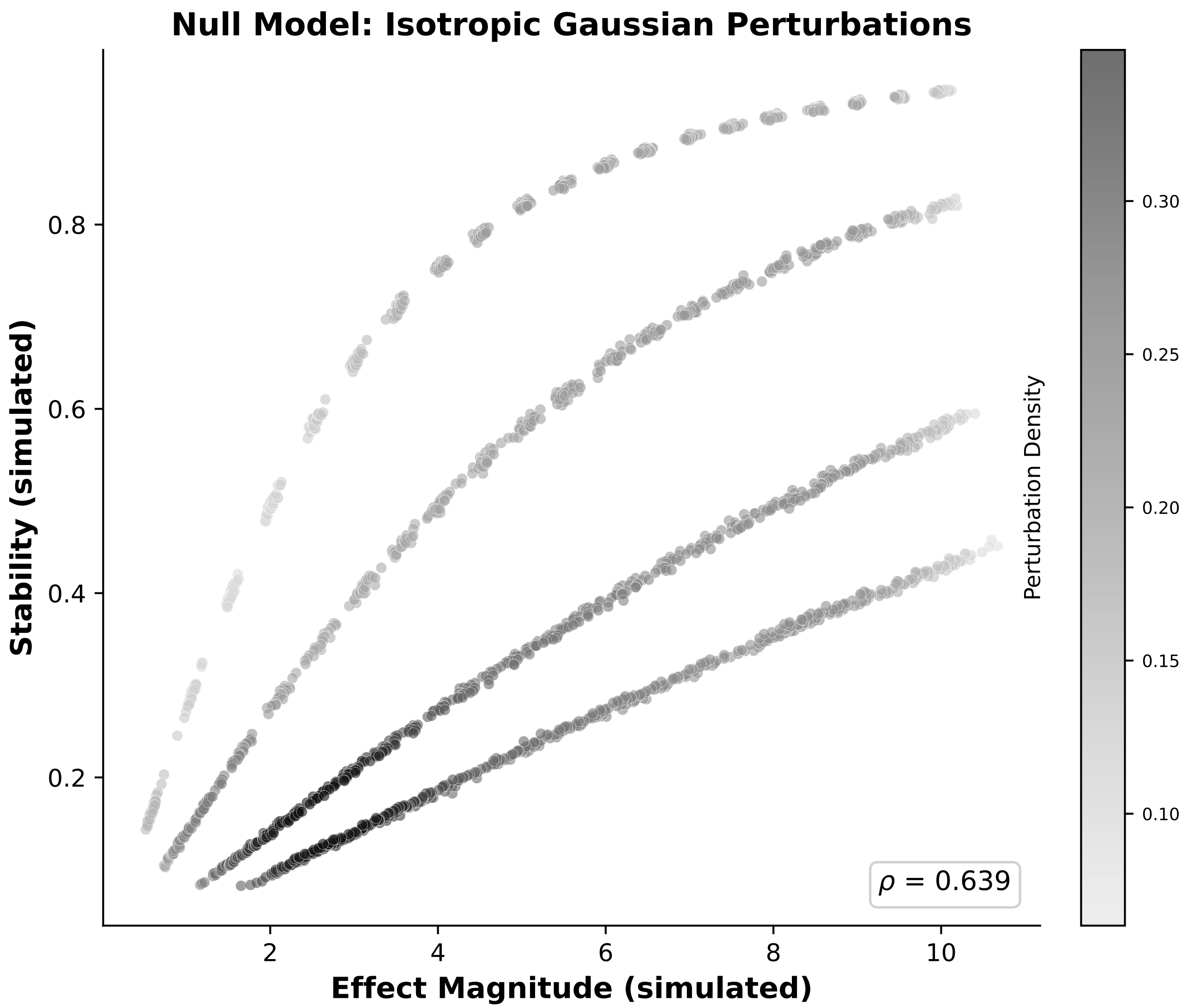
Supplementary Figure S5: **PCA dimensionality ablation.** Magnitude-stability Spearman ρ as a function of the number of principal components retained (10, 20, 30, 50, 100). Shaded regions indicate 95% bootstrap CIs (10,000 iterations). Norman 2019 shows stable correlations (ρ = 0.94–0.96) with overlapping CIs across all settings. Dixit 2016 shows modest improvement with more PCs (ρ = 0.67 to 0.79), suggesting higher-dimensional structure contributes to the relationship in that dataset. Replogle 2022 shows consistently high correlations (ρ = 0.95–0.98). Rank-order perturbation consistency is high across all settings (Norman: $r = 0.98 \pm 0.02$; Dixit: $r = 0.96 \pm 0.04$), confirming that the choice of 50 PCs does not drive the results.



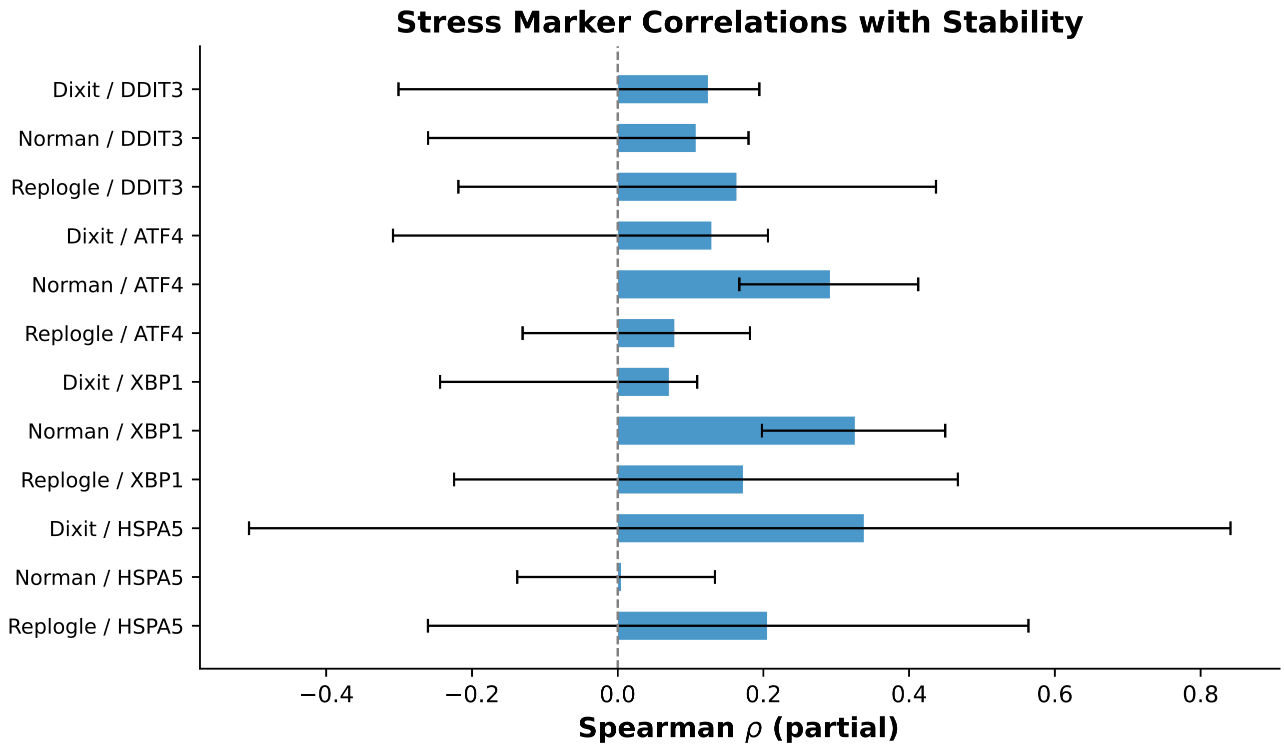
Supplementary Figure S6: **Random seed reproducibility.** Magnitude-stability Spearman ρ recomputed across 15 different random seeds ($\{3, 7, 9, 11, 12, 18, 103, 108, 320, 411, 724, 1754, 1991, 2222, 7258\}$) for Norman 2019 and Replogle 2022. All correlations are identical to machine precision (cross-seed $r = 1.000$), confirming that stochastic elements in the preprocessing pipeline (PCA initialization) have no effect on the final results.



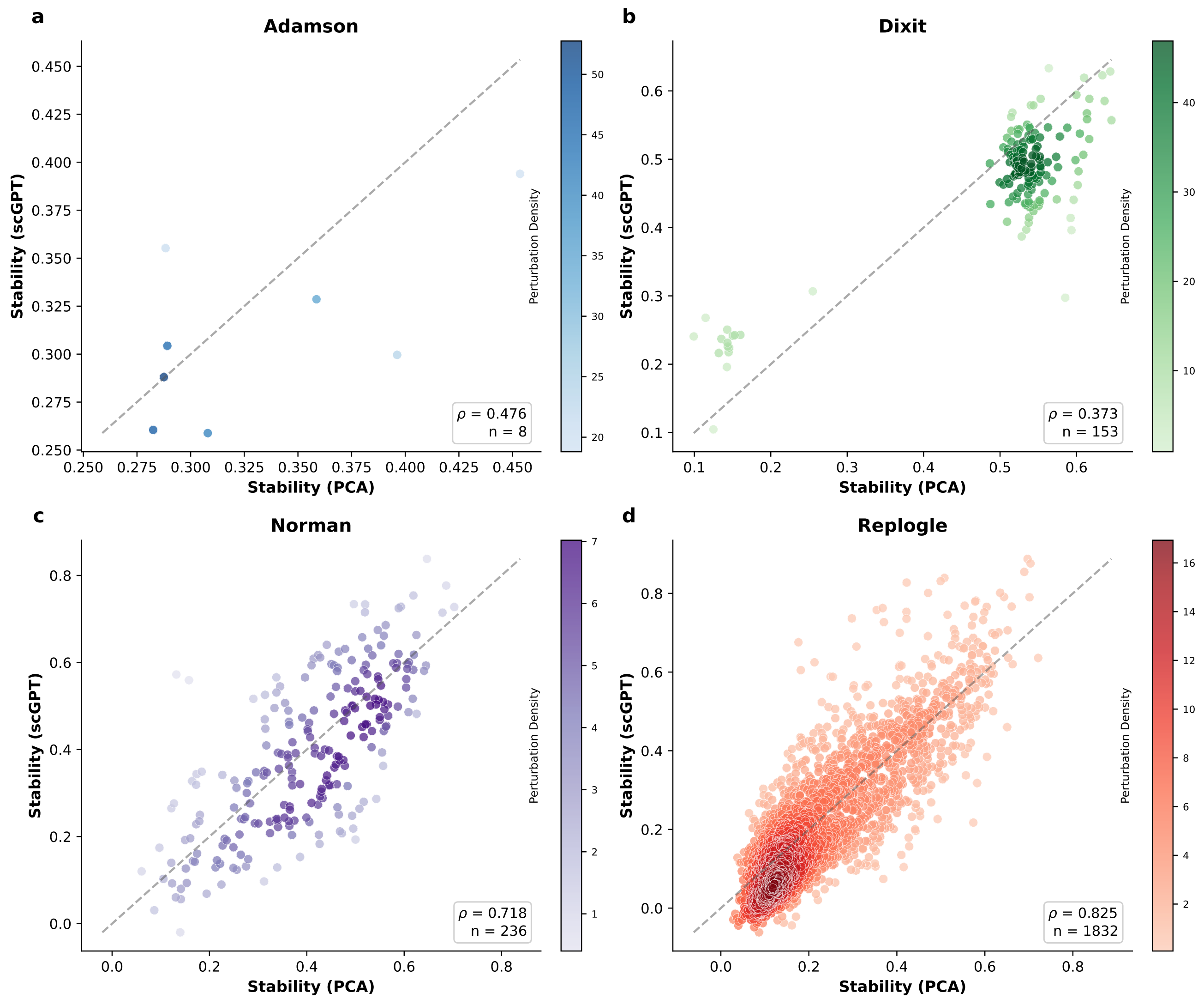
Supplementary Figure S7: **Leave-one-out influence analysis.** Distribution of $\Delta\rho$ values when each perturbation is removed in turn. The LOO range is narrow for all datasets: removing any single perturbation changes the correlation by at most $\Delta\rho = 0.002$ (Norman), $\Delta\rho = 0.014$ (Dixit), or $\Delta\rho < 0.001$ (Replogle). Most influential perturbations: BAK1 (most helpful, Norman), HES7 (most harmful, Norman), ELK1 (most helpful, Dixit), CREB1+E2F4+ELF1 (most harmful, Dixit).



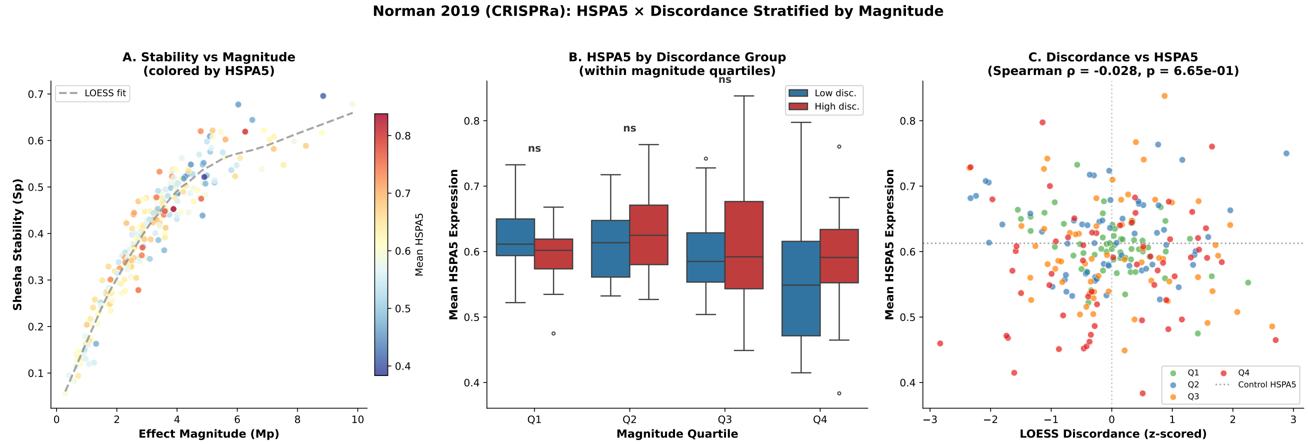
Supplementary Figure S8: **Theoretical null model under isotropic Gaussian perturbations.** Magnitude (x-axis) versus stability (y-axis) for 2,000 simulated perturbations ($d = 50$ dimensions, $\sigma \in \{0.5, 1.0, 2.0, 3.0\}$, 500 simulations per condition). Under the null model, stability is almost perfectly predicted by SNR ($\rho = 0.999$), with a partial correlation of $\rho_{\text{partial}} = 0.292$ after controlling for SNR. The heterogeneity observed in real data (Norman $\rho_{\text{partial}} = -0.859$, Dixit $\rho_{\text{partial}} = +0.627$) far exceeds this null prediction, confirming that biological factors beyond simple SNR confounding drive the magnitude-stability relationship.



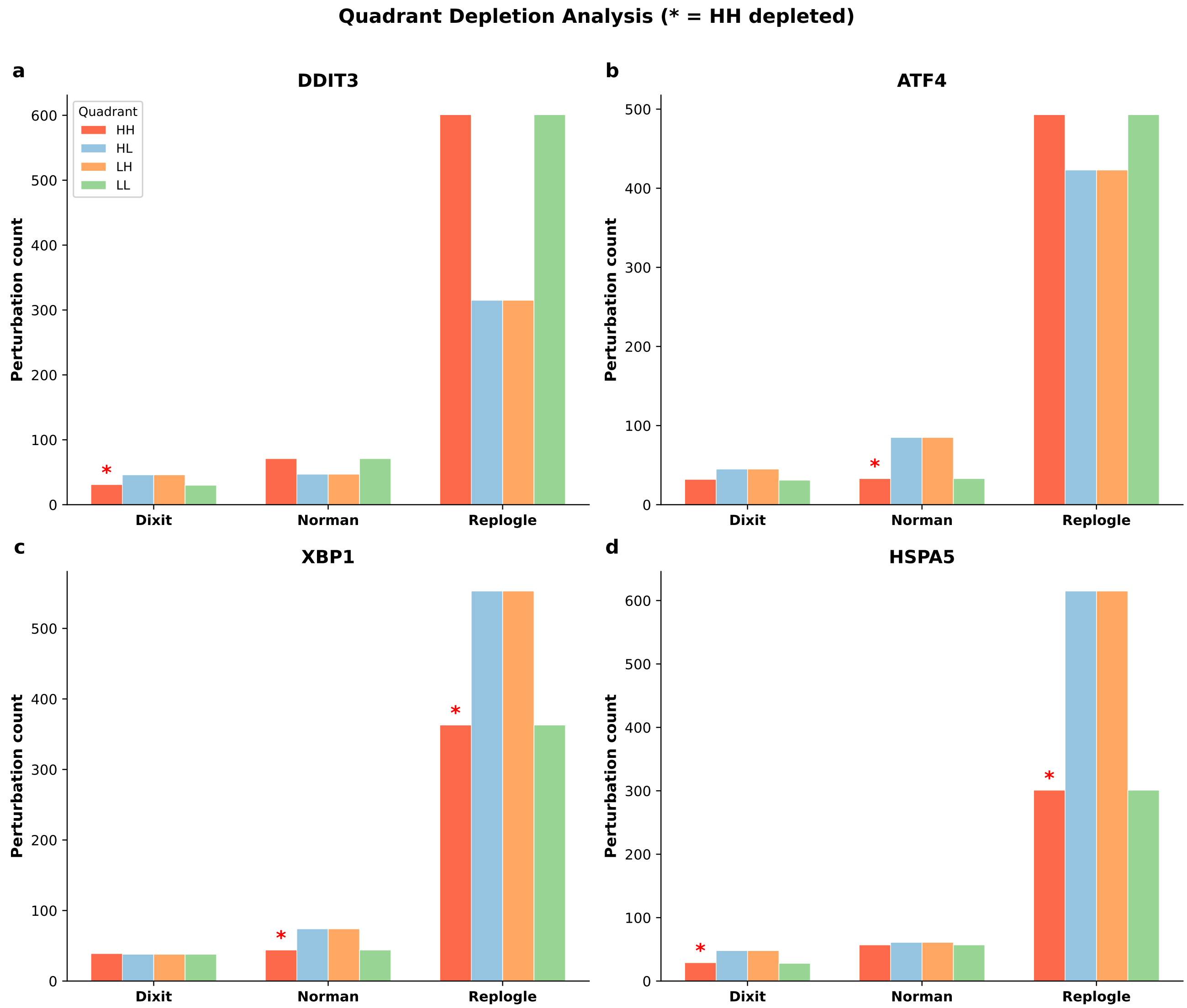
Supplementary Figure S9: **Stress marker correlations with geometric stability.** Forest plot of Spearman correlations between perturbation stability (S_p) and mean expression of four canonical stress response markers (DDIT3, ATF4, XBP1, HSPA5) across three datasets (Norman, Dixit, Replogle). Bars extend to 95% bootstrap CIs. Significant associations ($p < 0.001$) in bold. HSPA5 shows the most consistent negative association across datasets ($\rho = -0.31$ to -0.40 in Dixit and Replogle), while DDIT3 shows sign heterogeneity between CRISPRa and CRISPRi modalities, reflecting the directional effect of activation versus interference on stress pathway engagement.



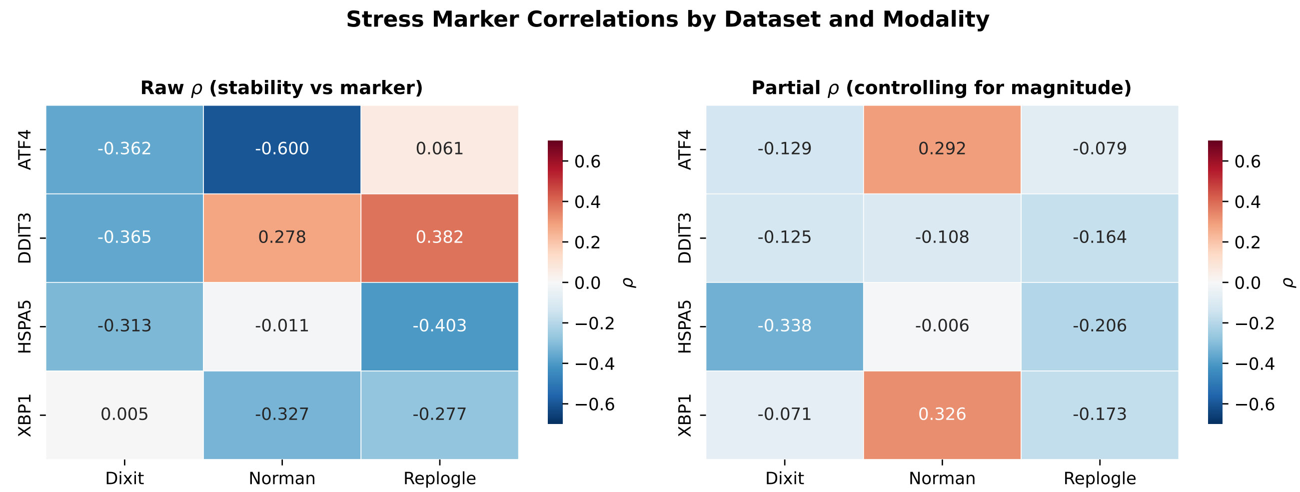
Supplementary Figure S10: **Per-perturbation concordance between PCA and scGPT stability estimates.** Each panel shows PCA-derived stability (x-axis) versus scGPT-derived stability (y-axis) for shared perturbations, with identity line (dashed). Point color indicates local perturbation density. Spearman ρ of paired values annotated. High concordance confirms that stability rankings are preserved across linear and nonlinear embedding spaces, and that the magnitude-stability relationship is a property of biological state space rather than an artifact of the dimensionality reduction method.



Supplementary Figure S11: **Magnitude-stratified HSPA5 analysis in Norman 2019 CRISPRa confirms attenuation rather than reversal of the stress-stability association.** (**A**) Stability versus magnitude for 236 Norman perturbations, colored by mean HSPA5 expression. Dashed line: LOESS fit. No visible gradient of HSPA5 with distance from the curve. (**B**) Mean HSPA5 expression for high-discordance (red) versus low-discordance (blue) perturbations within each magnitude quartile ($n = 59$ per bin, split at within-bin median LOESS discordance). Three of four quartiles show directionally higher HSPA5 in high-discordance perturbations (Q2: $\Delta = +0.015$; Q3: $+0.013$; Q4: $+0.027$, $p = 0.070$), but none reach significance. Q1 shows a small reversal ($\Delta = -0.021$). The pattern is consistent with a weak signal diluted by limited power rather than a true absence. (**C**) LOESS-residual discordance versus mean HSPA5, colored by magnitude quartile. The overall correlation is negligible (Spearman $\rho = -0.028$, $p = 0.67$), confirming the null partial correlation reported in the main text. Dotted horizontal line: control cell mean HSPA5. This analysis was performed at a reviewer's suggestion to test whether the CRISPRa null in Norman masks a within-quartile signal. The result supports the interpretation that the UPR-stability association is attenuated in CRISPRa rather than reversed, but is underpowered at $n = 59$ per bin to detect the modest effect sizes observed in CRISPRi datasets (4–5% of residual variance).



Supplementary Figure S12: **Quadrant depletion analysis of stability versus stress.** Perturbations split at median stability and median stress expression (dashed lines). Quadrant counts annotated. The high-stability/high-stress (HH) quadrant is systematically depleted across multiple stress markers and datasets (Fisher's exact test; Table S20), supporting the interpretation that geometric coherence is a prerequisite for cellular homeostasis. Perturbations producing incoherent cellular responses (low stability) are more likely to induce elevated stress signatures.



Supplementary Figure S13: **Stress marker correlations by dataset and modality.** Heatmap of Spearman correlations between geometric stability and four stress markers (DDIT3, ATF4, XBP1, HSPA5) across three datasets. Color scale: blue (positive) to red (negative), centered at zero. The heterogeneity across markers and datasets reflects differences in baseline stress levels, perturbation modality (CRISPRa versus CRISPRi), and the specific stress pathway engaged by each class of perturbation.

Table S1: Dataset overview.

Dataset	Modality	Cell type	Perturbations	Total cells	Median cells/pert	Control label
Norman 2019 (Norman et al. 2019)	CRISPRa	K562	236	111,255	352	control
Adamson 2016 (Adamson et al. 2016)	CRISPRi	K562	8	5,752	560	control
Dixit 2016 (Dixit et al. 2016)	CRISPRi	BMDCs	153	99,722	75	control
Papalexi 2021(Papalexi et al. 2021)	Pooled	THP-1	25	18,343	662	NT (pooled)
Replogle 2022 (Replogle et al. 2022)	CRISPRi	K562	1,832	310,385	132	non-targeting

Table S2: Mixed-effects model results.

Parameter	β	95% CI
Magnitude	0.168	[0.166, 0.170]
Spread	−0.122	[−0.128, −0.116]
Sample size	−0.015	[−0.017, −0.013]
Dataset variance	≈ 0	—

Table S3: Magnitude-stability Spearman correlations with 95% bootstrap CIs (10,000 iterations).

Dataset	n	ρ	95% CI	p
Norman	236	0.953	[0.934, 0.965]	$< 10^{-100}$
Adamson	8	0.929	[0.407, 1.000]	$< 10^{-4}$
Dixit	153	0.746	[0.641, 0.827]	$< 10^{-28}$
Replogle	1832	0.970	[0.966, 0.972]	$< 10^{-100}$
Papalexi	25	0.985	[0.939, 0.997]	$< 10^{-19}$

Table S4: Magnitude-stability correlation across distance metrics.

Dataset	$\rho_{\text{Euclidean}}$	ρ_{Whitened}	$\rho_{\text{K-nn}}$
Adamson	0.929 [0.407, 1.000]	0.976 [0.730, 1.000]	0.929 [0.522, 1.000]
Dixit	0.746 [0.641, 0.827]	0.965 [0.946, 0.975]	0.949 [0.921, 0.966]
Norman	0.953 [0.934, 0.965]	0.963 [0.947, 0.974]	0.951 [0.930, 0.964]
Replogle	0.970 [0.966, 0.972]	0.983 [0.980, 0.985]	0.978 [0.975, 0.981]
Papalexi	0.981 [0.927, 0.995]	0.988 [0.939, 0.999]	0.988 [0.937, 0.999]

Table S5: Spearman correlations between effect magnitude (M_p) and Shesha perturbation stability (S_p) across five CRISPR datasets, accompanying the LOESS fit comparison in Fig. S1. These rank correlations are fit-independent; the accompanying figure shows where linear and LOESS fits diverge despite identical ρ values. Bootstrap 95% CIs (10,000 iterations, seed 320, percentile method).

Dataset	n	ρ	**95% CI**	p
Norman 2019 (CRISPRa)	236	+0.959	[0.942, 0.970]	2.3×10^{-130}
Adamson 2016 (CRISPRi)	8	+0.929	[0.407, 1.000]	8.6×10^{-4}
Dixit 2016 (CRISPRi)	98	+0.788	[0.661, 0.876]	6.6×10^{-22}
Papalexi 2021 (Pooled)	24	+0.977	[0.912, 0.995]	3.7×10^{-16}
Replogle 2022 (CRISPRi)	1,832	+0.970	[0.966, 0.972]	$< 10^{-300}$
Pooled (z-scored)	2,198	+0.970	[0.967, 0.973]	$< 10^{-300}$

Table S6: Cross-method agreement for discordance rankings. Spearman correlations between three discordance methods (linear OLS residual, rank-based, LOESS with bandwidth fraction = 0.3) and top-10 overlap (number of shared genes in the top 10 most discordant). Agreement is highest in Dixit (narrow magnitude range) and lowest in Norman and Replogle (wide magnitude range where the nonlinearity is strongest).

		Spearman ρ			**Top-10 overlap**		
Dataset	n	Lin–LOESS	Lin–Rank	Rank–LOESS	Lin–LOESS	Lin–Rank	Rank–LOESS
Dixit 2016	153	0.918	0.895	0.832	8/10	4/10	2/10
Norman 2019	236	0.693	0.755	0.940	0/10	0/10	7/10
Papalexi 2021	25	0.722	0.710	0.691	8/10	6/10	6/10
Replogle 2022	1,832	0.810	0.776	0.925	1/10	0/10	3/10

In Norman and Replogle, the linear method disagrees substantially with both rank-based and LOESS methods (top-10 overlap $\leq 1/10$), while rank-based and LOESS methods agree well (overlap 7/10 and 3/10 respectively; overall $\rho = 0.940$ and 0.925). This pattern reflects the nonlinearity of the magnitude-stability relationship: the linear method systematically overestimates discordance for high-magnitude perturbations, a bias that the rank-based and LOESS methods correct. In Dixit, where the magnitude range is narrow (0.5–2.3), all three methods produce similar rankings.

Table S7: Top 10 discordant and top 5 concordant perturbations in Norman 2019 (PCA space), with linear and LOESS discordance. CEBPA+JUN, the most discordant perturbation under linear residuals (rank 1, disc. = +1.84), drops to rank 172 under LOESS with slightly negative discordance (−0.47). KLF1 is the most concordant single-gene perturbation under both methods.

Perturbation	M_p	S_p	n **cells**	**LOESS** Disc.	**LOESS** Rank	**Linear** Disc.	**Linear** Rank
Most discordant (LOESS ranking)							
PLK4+STIL	2.72	0.278	81	+2.88	1	+0.52	21
HES7	4.85	0.439	126	+2.70	2	+0.63	17
C3orf72+FOXL2	3.58	0.364	59	+2.65	3	+0.42	32
STIL	1.24	0.123	306	+2.26	4	+0.77	11
CBL+UBASH3A	3.10	0.345	64	+2.07	5	+0.27	56
FOXA3+FOXF1	3.60	0.391	175	+1.96	6	+0.24	60
CITED1	2.64	0.305	215	+1.95	7	+0.30	52
ELMSAN1+ZBTB10	4.38	0.445	182	+1.82	8	+0.31	48
MIDN	3.00	0.348	298	+1.75	9	+0.20	63
HOXC13	3.59	0.400	384	+1.72	10	+0.18	70
Most concordant (LOESS ranking)							
ETS2+MAPK1	6.03	0.677	459	−2.84	236	−0.36	198
KLF1	2.71	0.476	1954	−2.52	235	−0.87	236
KLF1+SET	3.42	0.533	666	−2.34	234	−0.86	235
DUSP9+KLF1	4.78	0.620	366	−2.33	233	−0.68	229
ELMSAN1+MAP2K3	2.32	0.422	581	−2.24	232	−0.72	232
CEBPA variants							
CEBPA+ZC3HAV1	7.54	0.548	86	+1.55	17	+1.42	4
CEBPA+CEBPB	8.28	0.589	64	+0.94	36	+1.57	3
CEBPA	8.81	0.617	458	+0.52	64	+1.68	2
CEBPA+CEBPE	7.88	0.598	179	+0.41	75	+1.27	5
CEBPA+KLF1	7.21	0.582	311	+0.38	81	+0.99	8
CEBPA+JUN	9.83	0.678	54	−0.47	172	+1.84	1

The CEBPA discordance is method-dependent. Under linear residuals, CEBPA variants occupy 5 of the top 8 discordant positions, driven by their high magnitudes (M_p = 7–10) in the regime where the linear fit underpredicts stability. Under LOESS, only CEBPA+ZC3HAV1 (rank 17) remains in the top 20; CEBPA+JUN drops from rank 1 to rank 172 with negative discordance. By contrast, KLF1 concordance is robust to method: KLF1 is rank 236 (linear) and rank 235 (LOESS), and all KLF1 combinations cluster at the concordant extreme under both methods.

Table S8: Top 5 discordant and top 5 concordant perturbations in Dixit 2016 (PCA space), with linear and LOESS discordance. Rankings are largely consistent across methods (top-10 overlap = 8/10) because the magnitude range is narrow ($M_p = 0.5$–4.6).

Perturbation	M_p	S_p	n **cells**	**LOESS** Disc.	**LOESS** Rank	**Linear** Disc.	**Linear** Rank
Most discordant (LOESS ranking)							
NR2C2+YY10	4.24	0.533	38	+3.03	1	+0.40	1
INTERGENIC393453+EGR1+ELF1	4.19	0.538	12	+2.20	2	+0.30	2
ETS1+YY10	4.34	0.563	35	+1.96	3	+0.27	4
CREB1+ELF1+ELK1	4.29	0.559	16	+1.85	4	+0.24	9
ELF1+ELK1+NR2C2	3.78	0.485	12	+1.75	5	+0.24	8
Most concordant (LOESS ranking)							
ELF1+ELF1+YY1	3.89	0.573	14	−3.06	153	−0.38	152
CREB1+ELF1+ELF1	3.98	0.580	10	−2.89	152	−0.33	148
CREB1+E2F4+ELF1	3.87	0.567	15	−2.82	151	−0.35	150
INTERGENIC1216445+YY1	3.97	0.572	34	−2.36	150	−0.26	143
CREB1+ELK1+ETS1	4.09	0.589	10	−2.19	149	−0.26	145

Rankings are highly consistent across methods, confirming that the nonlinear correction primarily affects datasets with wide magnitude ranges (Norman, Replogle). Cell counts are low for several Dixit perturbations ($n < 15$), reflecting the smaller scale of this early Perturb-seq experiment.

Table S9: Top 10 discordant and top 5 concordant perturbations in Replogle 2022 (PCA space). Discordance is computed by two methods: linear (standardized OLS residual) and LOESS (sign-flipped locally weighted residual, bandwidth fraction = 0.3). Positive values indicate lower stability than predicted by magnitude. Rankings differ substantially for high-magnitude perturbations where the nonlinearity in the magnitude-stability relationship is strongest. LOESS rankings are used throughout the main text.

Gene	M_p	S_p	n **cells**	**LOESS** Disc.	**LOESS** Rank	**Linear** Disc.	**Linear** Rank	**Function**
Most discordant (LOESS ranking)								
CHMP2A	3.01	0.183	93	+5.07	1	+1.05	21	ESCRT-III complex
SF3B3	3.37	0.222	381	+4.83	2	+1.01	23	U2 snRNP (spliceosome)
SF3B2	3.35	0.231	500	+4.55	3	+0.94	26	U2 snRNP (spliceosome)
PSMD7	4.99	0.356	373	+4.23	4	+1.15	15	26S proteasome (19S cap)
CHMP3	8.36	0.534	130	+4.20	5	+2.14	2	ESCRT-III complex
SMU1	4.71	0.341	155	+4.18	6	+1.08	19	Spliceosome-assoc. factor
COPB2	2.88	0.203	129	+4.17	7	+0.82	33	COPI vesicle coat
CRNKL1	2.21	0.139	261	+4.02	8	+0.83	31	NTC/Prp19 (spliceosome)
TSG101	5.48	0.395	87	+3.99	9	+1.21	11	ESCRT-I complex
NSRP1	3.86	0.293	237	+3.88	10	+0.85	29	Splicing regulatory protein
Most concordant (LOESS ranking)								
CASP8AP2	5.31	0.651	211	−3.19	1832	−0.63	1823	Histone mRNA proc. (FLASH)
CHAF1B	4.91	0.605	84	−2.61	1831	−0.57	1811	Chromatin assembly factor 1
LSG1	3.48	0.499	116	−2.43	1830	−0.79	1832	Ribosome biogenesis
NOL6	4.84	0.592	114	−2.39	1829	−0.54	1800	Ribosome biogenesis
ZFC3H1	4.60	0.574	117	−2.31	1828	−0.57	1807	PAXT complex (RNA decay)
Linear flagships under LOESS correction								
GATA1	10.08	0.697	108	+2.02	89	+2.15	1	Erythroid/megakaryocytic TF
AQR	9.19	0.663	90	+1.79	105	+1.80	3	RNA helicase (spliceosome)

Five of the ten LOESS-discordant genes are spliceosome components, three are ESCRT/membrane remodeling factors, and one is a proteasome subunit. All encode subunits of large, essential multi-protein complexes. GATA1 and AQR, which ranked 1st and 3rd under linear residuals, drop to ranks 89 and 105 under LOESS, indicating that their apparent discordance was amplified by the nonlinearity at high magnitudes. CHMP3 (ESCRT-III) is the only gene from the original linear top 3 that retains a top-5 ranking under LOESS. Among the concordant genes, CASP8AP2 (histone mRNA processing) and CHAF1B (chromatin assembly) join ribosome biogenesis factors LSG1 and NOL6 at the coherent extreme.

Table S10: Three-tier comparison of S_p versus perturbation-response score (PS) in Replogle 2022 ($n = 1{,}832$). PS was computed at three levels of fidelity: Tier 1 (mean per-cell Euclidean distance from control centroid), Tier 2 (mean per-cell Mahalanobis distance, re-weighted by inverse control covariance), and Tier 3 (Python port of Song et al.'s scMAGeCK constrained-optimization algorithm). The centroid-based proxies (Tiers 1–2) correlate weakly with the algorithm-derived PS ($\rho < 0.15$), indicating that per-cell distance from the control centroid does not approximate Song et al.'s score. Bootstrap 95% CIs (10,000 iterations, seed 320).

		Tier 1 Euclidean	**Tier 2** Mahalanobis	**Tier 3** scMAGeCK
Raw correlations with PS				
S_p vs PS	ρ	+0.764	+0.802	+0.447
M_p vs PS	ρ	+0.876	+0.896	+0.327
Partial correlation (S_p vs PS | magnitude)				
	ρ_{partial}	−0.883	−0.596	+0.507
	95% CI	[−0.894, −0.870]	[−0.637, −0.552]	[+0.468, +0.544]
	Shared variance	58.4%	64.4%	20.0%
Inter-tier correlations				
Euclid vs Mahal	ρ	+0.972		
Euclid vs Real	ρ			+0.097
Mahal vs Real	ρ			+0.149
Incremental UPR prediction				
$S_p \mid M_p \rightarrow$ UPR	ρ	−0.215	($p = 1.2 \times 10^{-20}$)	
$S_p \mid M_p$+PS $\rightarrow$ UPR	ρ	−0.139	−0.284	−0.203
	p	2.5×10^{-9}	2.5×10^{-35}	1.9×10^{-18}
	Sp adds?	Yes	Yes	Yes
PS $\mid M_p$+$S_p \rightarrow$ UPR	ρ	−0.045	−0.232	−0.072
	p	0.054	8.0×10^{-24}	2.1×10^{-3}
	PS adds?	No	Yes	No

The partial correlation between S_p and PS reverses sign across tiers: strongly negative with centroid-based proxies (Tiers 1–2), moderately positive with the algorithm-derived PS (Tier 3). This reversal reflects the fact that centroid-based distance is a noisy magnitude proxy (M_p-PS $\rho > 0.87$) rather than a measure of per-cell response strength. Critically, S_p's incremental prediction of UPR pathway activation is robust across all three tiers ($\rho = -0.14$ to -0.28, all $p < 10^{-9}$), confirming that directional coherence captures stress-relevant information beyond both magnitude and per-cell response strength regardless of how PS is operationalized.

Table S11: PCA dimensionality ablation. Magnitude-stability correlations (ρ) with 95% bootstrap CIs (10,000 iterations) across varying numbers of principal components.

Dataset	PCs	ρ	95% CI	p
Norman	10	0.946	[0.928, 0.958]	$< 10^{-116}$
	20	0.942	[0.921, 0.956]	$< 10^{-113}$
	30	0.945	[0.926, 0.958]	$< 10^{-116}$
	50	0.953	[0.935, 0.965]	$< 10^{-123}$
	100	0.964	[0.949, 0.973]	$< 10^{-137}$
Dixit	10	0.669	[0.547, 0.767]	$< 10^{-21}$
	20	0.685	[0.558, 0.786]	$< 10^{-22}$
	30	0.700	[0.582, 0.795]	$< 10^{-24}$
	50	0.746	[0.642, 0.828]	$< 10^{-28}$
	100	0.793	[0.703, 0.862]	$< 10^{-34}$
Replogle	10	0.947	[0.941, 0.952]	$< 10^{-300}$
	20	0.961	[0.956, 0.964]	$< 10^{-300}$
	30	0.965	[0.961, 0.968]	$< 10^{-300}$
	50	0.970	[0.966, 0.973]	$< 10^{-300}$
	100	0.976	[0.973, 0.979]	$< 10^{-300}$

Table S12: Random seed reproducibility. Stability recomputed using 15 different random seeds per dataset. All correlations are identical to machine precision (cross-seed $r = 1.000$), confirming no stochastic dependence in the analytic pipeline.

Dataset	n	Seeds	ρ (mean ± std)	95% CI	Cross-seed r
Norman	236	15	0.945403 ± 0.000000	[0.925, 0.959]	1.000
Replogle	1832	15	0.965106 ± 0.000000	[0.961, 0.969]	1.000

Table S13: Leave-one-out influence analysis. Removing any single perturbation changes the correlation by at most $\Delta\rho = 0.014$ (Dixit) or $\Delta\rho = 0.002$ (Norman), confirming no individual perturbation drives the observed relationship.

Dataset	n	Full ρ	LOO range	Most helpful ($\Delta\rho$)	Most harmful ($\Delta\rho$)
Norman	236	0.945	[0.945, 0.948]	BAK1 (+0.0007)	HES7 (−0.0024)
Dixit	153	0.700	[0.695, 0.714]	ELK1 (+0.0060)	CREB1+E2F4+ELF1 (−0.0139)
Replogle	1832	0.965	[0.965, 0.965]	EIF2S1 (+0.0001)	CRNKL1 (−0.0004)

Table S14: Partial correlations between stability and magnitude, controlling for intrinsic spread and sample size. The heterogeneity across datasets (Norman $\rho_{\text{partial}} = -0.86$, Dixit $\rho_{\text{partial}} = +0.63$) exceeds what the isotropic null model predicts, indicating biological factors beyond SNR confounding.

Dataset	ρ_{partial}	95% CI	p
Norman	−0.859	[−0.905, −0.781]	$< 10^{-70}$
Dixit	0.627	[0.482, 0.728]	$< 10^{-18}$
Replogle	−0.789	[−0.812, −0.765]	$< 10^{-300}$
Pooled	−0.102	[−0.156, −0.049]	$< 10^{-6}$

The sign reversal between Norman ($\rho_{\text{partial}} = -0.859$) and Dixit (+0.627) likely reflects biological differences between the datasets: Dixit perturbations are applied to LPS-stimulated BMDCs with high baseline heterogeneity, where stronger perturbations may produce more coherent responses by overriding the noisy activation state. The paper's main claims (discordance, UPR association, reproducibility prediction) are computed from magnitude-controlled analyses that do not require this partial correlation to be sign-consistent.

Table S15: Spearman correlation of Sp-discordance with alternative perturbation heterogeneity metrics. Discordance is the standardized residual from the magnitude-stability OLS regression (positive values indicate lower stability than predicted by magnitude). Six alternative metrics are compared: within-perturbation spread (mean variance across PCs), coefficient of variation of per-cell L_2 norms, number of differentially expressed genes ($|\text{LFC}| > 0.5$, $\text{FDR} < 0.05$, t-test), mean absolute log-fold-change across all tested genes, Nadig-style η^2 (proportion of PCA variance explained by perturbation identity, averaged across PCs), and Song PS proxy (mean per-cell Euclidean distance from control centroid). 95% bootstrap CIs (2,000 iterations, seed 320).

Metric	**Norman 2019** (n = 132)		**Replogle 2022** (n = 1,832)	
	ρ	95% CI	ρ	95% CI
Geometric heterogeneity (single-cell level)				
CV of cell L_2 norms	+0.678	[+0.546, +0.785]	+0.574	[+0.537, +0.608]
Within-pert. spread (mean PC variance)	+0.519	[+0.364, +0.651]	+0.412	[+0.369, +0.451]
Song PS proxy (mean cell distance)	+0.250	[+0.065, +0.418]	+0.120	[+0.074, +0.169]
Transcriptomic breadth (pseudobulk / gene level)				
Nadig-style η^2 (variance decomposition)	−0.199	[−0.363, −0.021]	−0.276	[−0.327, −0.226]
Mean |LFC|	+0.185	[+0.009, +0.368]	−0.270	[−0.318, −0.221]
DE gene count (|LFC| > 0.5, FDR < 0.05)	−0.054	[−0.257, +0.141]	−0.219	[−0.269, −0.169]

Metrics are grouped by the level at which they operate. Geometric heterogeneity metrics (top) quantify single-cell-level variation and correlate positively with Sp-discordance: perturbations that are geometrically incoherent produce cells with more variable distances from control (high CV) and greater within-perturbation spread. Transcriptomic breadth metrics (bottom) quantify gene-level effects via pseudobulk differential expression and correlate negatively with Sp-discordance: geometrically incoherent perturbations explain *less* variance in aggregate (η^2) and produce fewer detectable DE genes, consistent with geometric scattering diluting the population-level signal that pseudobulk methods depend on. The Song PS proxy shows weak positive correlation with discordance, indicating near-orthogonality between per-cell response magnitude and geometric incoherence at the discordance level (the stronger anticorrelation between S_p and PS reported in the main text reflects the relationship after controlling for magnitude, which is not captured by discordance). Mean |LFC| shows a sign reversal between Norman (+0.185) and Replogle (−0.270), possibly reflecting differences in perturbation scale or cell type context.

Table S16: Functional diversity of differentially expressed genes versus Sp-discordance in Norman 2019. For each perturbation, the top-k genes (ranked by absolute log-fold change versus control) were annotated with GO Biological Process terms via g:Profiler (FDR < 0.05). Functional diversity is the number of distinct GO:BP terms enriched. Correlations are reported for both linear and LOESS discordance at three values of k to assess sensitivity. Perturbations with zero enriched GO:BP terms were excluded from correlation analyses. Partial correlations control for effect magnitude.

k	**Method**	n	ρ	**95% CI**	p	ρ_{partial}	p_{partial}
25	Linear	130	+0.165	$[-0.019, +0.338]$	6.1×10^{-2}	+0.180	4.1×10^{-2}
25	LOESS	130	−0.098	$[-0.267, +0.079]$	2.7×10^{-1}	−0.117	1.9×10^{-1}
50	Linear	185	−0.015	$[-0.162, +0.133]$	8.4×10^{-1}	+0.017	8.2×10^{-1}
50	LOESS	185	−0.119	$[-0.255, +0.022]$	1.1×10^{-1}	−0.113	1.3×10^{-1}
100	Linear	217	+0.058	$[-0.086, +0.203]$	4.0×10^{-1}	+0.105	1.2×10^{-1}
100	LOESS	217	−0.034	$[-0.168, +0.102]$	6.2×10^{-1}	−0.026	7.0×10^{-1}

Group	n	**Mean GO:BP**	**Median**	**Mean disc. (LOESS)**
CEBP family	20	21.1	22	+0.036
KLF1 combinations	10	9.6	7	−1.380
Mann-Whitney $U = 151$, one-sided $p = 0.013$ (CEBP > KLF1)				

The global discordance-diversity correlation is not significant under any combination of k and residual method, indicating that transcriptional target diversity does not fully explain geometric incoherence. However, the targeted CEBP versus KLF1 comparison is significant ($p = 0.013$), confirming that the known pleiotropic regulator engages more diverse biological processes than the known lineage-specific regulator. The sign inconsistency between linear ($\rho > 0$ at $k = 25, 100$) and LOESS ($\rho < 0$) reflects the sensitivity of these correlations to the residual method at the modest effect sizes observed.

Table S17: Split-half reproducibility benchmarking. For each perturbation, cells were randomly partitioned into two equal halves (50 independent splits, seed 320), and the cosine similarity between half-shift vectors was averaged across splits. Panel A: partial correlations of each predictor with split-half reproducibility controlling for magnitude, across three PS implementations and two datasets. Panel B: magnitude-matched quartile analysis for Replogle, comparing high-S_p versus low-S_p perturbations within each magnitude bin. PS was computed at three tiers: Euclidean (mean per-cell distance from control centroid), Mahalanobis (re-weighted by inverse control covariance), and Real (Python port of Song et al.'s scMAGeCK algorithm).

Panel A: Predictors of split-half reproducibility (partial ρ, controlling for magnitude)

	Replogle (n = 1,832)		**Norman** (n = 236)	
Predictor	Partial ρ	p	Partial ρ	p
S_p (stability)	+0.387	1.3×10^{-66}	+0.485	2.6×10^{-15}
PS (Euclidean proxy)	−0.249	3.5×10^{-27}	−0.504	1.4×10^{-16}
PS (Mahalanobis proxy)	−0.184	2.0×10^{-15}	−0.288	6.7×10^{-6}
PS (scMAGeCK algorithm)	−0.193	7.1×10^{-17}	−0.180	5.6×10^{-3}
Residualized AUC (predicting top-quartile reproducibility after removing magnitude)				
S_p | magnitude	0.539		0.716	
PS (scMAGeCK) | magnitude	0.448		0.592	

Panel B: Magnitude-matched quartile analysis (Replogle)

Quartile	**Mag. range**	**High-S_p** mean cos	**Low-S_p** mean cos	Δ	p
Q1 (lowest)	[0.35, 0.87]	0.243	0.137	+0.106	1.3×10^{-17}
Q2	[0.87, 1.55]	0.603	0.445	+0.159	4.1×10^{-23}
Q3	[1.55, 2.88]	0.847	0.790	+0.057	3.3×10^{-10}
Q4 (highest)	[2.89, 10.08]	0.967	0.936	+0.032	3.2×10^{-28}

Panel C: Magnitude-matched quartile analysis (Norman)

Quartile	**Mag. range**	**High-S_p** mean cos	**Low-S_p** mean cos	Δ	p
Q1 (lowest)	[0.29, 1.87]	0.940	0.784	+0.156	5.8×10^{-16}
Q2	[1.93, 2.98]	0.976	0.966	+0.009	5.2×10^{-3}
Q3	[3.00, 4.02]	0.984	0.969	+0.015	1.0×10^{-6}
Q4 (highest)	[4.03, 9.83]	0.991	0.982	+0.009	1.6×10^{-5}

S_p shows a consistent positive partial correlation with reproducibility across both datasets and all magnitude strata. All three PS implementations show the opposite: negative partial correlations, indicating that at equivalent magnitude, higher per-cell response strength predicts *lower* directional reproducibility. This has a geometric interpretation: perturbations where individual cells travel farther from control but in heterogeneous directions produce large per-cell distances (high PS) but poor population-level reproducibility. The S_p advantage is largest at low magnitudes (Q1–Q2), where screen hit prioritization is most difficult. Norman shows higher baseline reproducibility than Replogle (median split-half cosine 0.974 vs 0.861), consistent with fewer perturbations and larger per-perturbation cell counts.

Table S18: Stress marker correlations with geometric stability. Spearman correlations between perturbation stability (S_p) and mean expression of canonical stress markers. 95% bootstrap CIs (10,000 iterations). Significant results ($p < 0.001$) in **bold**.

Dataset	**DDIT3**		**ATF4**		**XBP1**		**HSPA5**	
	ρ	CI	ρ	CI	ρ	CI	ρ	CI
Norman	**0.278**	[0.16, 0.39]	**−0.600**	[−0.68, −0.51]	**−0.327**	[−0.44, −0.20]	−0.011	[−0.14, 0.12]
Dixit	**−0.365**	[−0.52, −0.20]	**−0.362**	[−0.52, −0.19]	0.005	[−0.15, 0.16]	**−0.313**	[−0.47, −0.15]
Replogle	**0.382**	[0.34, 0.43]	0.061	[0.01, 0.11]	**−0.277**	[−0.33, −0.23]	**−0.403**	[−0.45, −0.36]

Table S19: PCA vs scGPT magnitude-stability correlations.

Dataset	PCA ρ	scGPT ρ	scGPT 95% CI	$\Delta\rho$
Norman 2019	0.953	0.935	[0.911, 0.951]	−0.018
Dixit 2016	0.746	0.712	[0.585, 0.818]	−0.034
Replogle 2022	0.970	0.851	[0.836, 0.865]	−0.119

Table S20: Quadrant depletion tests. Perturbations split at median stability and median stress expression. The high-stability/high-stress (HH) quadrant is tested for depletion via Fisher's exact test. Significant depletions (**bold**) suggest geometric coherence is a prerequisite for cellular homeostasis.

Dataset	Marker	n	HH	HL	LH	LL	Fisher p	Depleted?
Dixit	**DDIT3**	**153**	**31**	**46**	**46**	**30**	**0.015**	**Yes**
Norman	DDIT3	236	71	47	47	71	0.003	No
Replogle	DDIT3	1832	601	315	315	601	$< 10^{-41}$	No
Dixit	ATF4	153	32	45	45	31	0.036	No
Norman	**ATF4**	**236**	**33**	**85**	**85**	**33**	$< 10^{-11}$	**Yes**
Replogle	ATF4	1832	493	423	423	493	0.001	No
Dixit	XBP1	153	39	38	38	38	1.000	No
Norman	**XBP1**	**236**	**44**	**74**	**74**	**44**	$< 10^{-4}$	**Yes**
Replogle	**XBP1**	**1832**	**363**	**553**	**553**	**363**	$< 10^{-19}$	**Yes**
Dixit	**HSPA5**	**153**	**29**	**48**	**48**	**28**	**0.002**	**Yes**
Norman	HSPA5	236	57	61	61	57	0.696	No
Replogle	**HSPA5**	**1832**	**301**	**615**	**615**	**301**	$< 10^{-49}$	**Yes**

Table S21: Pathway-level stress correlations with geometric stability. Spearman correlations (raw and partial, controlling for effect magnitude) between perturbation stability (S_p) and composite pathway scores computed via `scanpy.tl.score_genes` using curated MSigDB Hallmark gene sets. 95% bootstrap CIs (10,000 iterations). Significant partial correlations after Benjamini-Hochberg correction ($q < 0.05$) in **bold**. Gene set overlap with each dataset's transcriptome is shown as k/N (genes detected / genes in set). Adamson 2016 ($n = 8$) excluded due to insufficient sample size.

Pathway	**Dataset**	n	**Overlap**	ρ_{raw}	ρ_{partial}	**95% CI**	p_{partial}
Unfolded Protein Response (HALLMARK_UPR)							
	Dixit 2016 (CRISPRi)	153	77/78	−0.088	**−0.231**	[−0.383, −0.064]	4.1×10^{-3}
	Norman 2019 (CRISPRa)	236	78/78	−0.181	−0.023	[−0.158, +0.117]	7.3×10^{-1}
	Papalexi 2021 (CRISPR)	25	77/78	−0.507	−0.395	[−0.814, +0.046]	5.1×10^{-2}
	Replogle 2022 (CRISPRi)	1,832	72/78	−0.128	**−0.214**	[−0.273, −0.158]	1.7×10^{-20}
Apoptosis (HALLMARK_APOPTOSIS)							
	Dixit 2016 (CRISPRi)	153	70/73	−0.180	−0.079	[−0.252, +0.102]	3.3×10^{-1}
	Norman 2019 (CRISPRa)	236	71/73	−0.262	**−0.324**	[−0.463, −0.163]	3.6×10^{-7}
	Papalexi 2021 (CRISPR)	25	72/73	−0.432	+0.232	[−0.328, +0.700]	2.6×10^{-1}
	Replogle 2022 (CRISPRi)	1,832	53/73	+0.094	**−0.333**	[−0.383, −0.282]	1.2×10^{-48}
p53 Pathway (HALLMARK_P53_PATHWAY)							
	Dixit 2016 (CRISPRi)	153	46/51	−0.246	+0.011	[−0.170, +0.186]	8.9×10^{-1}
	Norman 2019 (CRISPRa)	236	48/51	+0.442	−0.129	[−0.268, +0.010]	4.9×10^{-2}
	Papalexi 2021 (CRISPR)	25	48/51	−0.287	+0.031	[−0.477, +0.597]	8.8×10^{-1}
	Replogle 2022 (CRISPRi)	1,832	32/51	+0.241	**−0.272**	[−0.319, −0.224]	1.6×10^{-32}
Reactive Oxygen Species (HALLMARK_ROS_PATHWAY)							
	Dixit 2016 (CRISPRi)	153	54/55	−0.128	**−0.276**	[−0.441, −0.104]	5.5×10^{-4}
	Norman 2019 (CRISPRa)	236	52/55	−0.339	+0.098	[−0.046, +0.242]	1.3×10^{-1}
	Papalexi 2021 (CRISPR)	25	54/55	−0.175	−0.058	[−0.602, +0.383]	7.8×10^{-1}
	Replogle 2022 (CRISPRi)	1,832	45/55	−0.127	−0.026	[−0.089, +0.039]	2.7×10^{-1}
Cross-dataset sign consistency of partial correlations							
UPR				All 4 datasets negative: **YES**			
Apoptosis				All 4 datasets negative: NO (Papalexi +0.232)			
p53				All 4 datasets negative: NO (Dixit +0.011, Papalexi +0.031)			
ROS				All 4 datasets negative: NO (Norman +0.098)			

Table S22: HSPA5 expression by discordance group within magnitude quartiles in Norman 2019 CRISPRa ($n = 236$). Perturbations in each magnitude quartile were split at the median LOESS-residual discordance. High-discordance perturbations show directionally higher HSPA5 in Q2–Q4 but no bin reaches $p < 0.05$, consistent with attenuation rather than absence of the UPR-stability association in CRISPRa.

Quartile	**Mp range**	n	**High-disc** HSPA5	**Low-disc** HSPA5	Δ	p (MWU)
Q1 (lowest)	[0.29, 1.87]	59	0.596	0.617	−0.021	0.971
Q2	[1.93, 2.98]	59	0.631	0.616	+0.015	0.184
Q3	[3.00, 4.02]	59	0.608	0.595	+0.013	0.260
Q4 (highest)	[4.03, 9.83]	59	0.587	0.559	+0.027	0.070

Table S23: Functional category stratification of the stability-UPR association in Replogle 2022. Partial Spearman correlations between perturbation stability (S_p) and UPR pathway score, controlling for effect magnitude, computed within functional categories defined by GO molecular function annotations. Categories with fewer than 15 perturbations were excluded (Chromatin Regulators, $n = 12$; Metabolic Enzymes, $n = 13$; Transcription Factors, $n = 11$). One gene (RPS27A) appeared in two categories (Ribosome Biogenesis and Proteasome/Ubiquitin). 95% bootstrap CIs (10,000 iterations). Benjamini-Hochberg correction applied across the six tested categories.

Category	n	ρ_{partial}	**95% CI**	p	**Sign**
ALL (global)	1,832	−0.214	[−0.274, −0.156]	1.7×10^{-20}	−
Per-category					
DNA Repair / Cell Cycle	35	−0.046	[−0.411, +0.326]	7.9×10^{-1}	−
ESCRT / Membrane Trafficking	26	+0.270	[−0.234, +0.631]	1.8×10^{-1}	+
Proteasome / Ubiquitin	39	−0.234	[−0.626, +0.138]	1.5×10^{-1}	−
Ribosome Biogenesis	84	−0.096	[−0.333, +0.136]	3.8×10^{-1}	−
Splicing / RNA Processing	37	+0.302	[−0.058, +0.557]	6.9×10^{-2}	+
Unassigned					
Remaining perturbations	1,576	−0.258	[−0.324, −0.193]	2.2×10^{-25}	−

Summary: 3/5 tested categories share the dominant negative sign. No individual category reaches significance after BH correction ($q < 0.05$: 1/6), indicating that the global association is not driven by any single functional class. The ESCRT sign reversal (+0.270) and Splicing reversal (+0.302) are non-significant and may reflect category-specific biology or limited power.

Table S24: Single-marker stress correlations with geometric stability: raw and partial. Spearman correlations between perturbation stability (S_p) and mean expression of four canonical stress response markers, with partial correlations controlling for effect magnitude. 95% bootstrap CIs (10,000 iterations). Partial correlations surviving magnitude control (CI excludes zero and $|\rho| > 0.1$) marked with †. Adamson 2016 excluded ($n = 8$).

Marker	**Dataset**	n	ρ_{raw}	**CI**	ρ_{partial}	**CI**	**Effect**
HSPA5 (BiP/GRP78)							
	Dixit 2016 (CRISPRi)	153	−0.313	[−0.464, −0.154]	−0.338†	[−0.506, −0.164]	med-large
	Norman 2019 (CRISPRa)	236	−0.011	[−0.141, +0.122]	−0.006	[−0.138, +0.133]	negligible
	Replogle 2022 (CRISPRi)	1,832	−0.403	[−0.448, −0.357]	−0.206†	[−0.260, −0.152]	sm-med
DDIT3 (CHOP)							
	Dixit 2016 (CRISPRi)	153	−0.365	[−0.517, −0.196]	−0.125	[−0.301, +0.055]	small
	Norman 2019 (CRISPRa)	236	+0.278	[+0.158, +0.394]	−0.108	[−0.260, +0.036]	small
	Replogle 2022 (CRISPRi)	1,832	+0.382	[+0.336, +0.425]	−0.164†	[−0.219, −0.109]	small
ATF4							
	Dixit 2016 (CRISPRi)	153	−0.362	[−0.519, −0.187]	−0.129	[−0.308, +0.053]	small
	Norman 2019 (CRISPRa)	236	−0.600	[−0.677, −0.508]	+0.292†	[+0.167, +0.413]	sm-med
	Replogle 2022 (CRISPRi)	1,832	+0.061	[+0.008, +0.113]	−0.079	[−0.130, −0.024]	negligible
XBP1							
	Dixit 2016 (CRISPRi)	153	+0.005	[−0.155, +0.162]	−0.071	[−0.244, +0.109]	negligible
	Norman 2019 (CRISPRa)	236	−0.327	[−0.441, −0.204]	+0.326†	[+0.198, +0.450]	med-large
	Replogle 2022 (CRISPRi)	1,832	−0.277	[−0.324, −0.227]	−0.173†	[−0.224, −0.121]	small